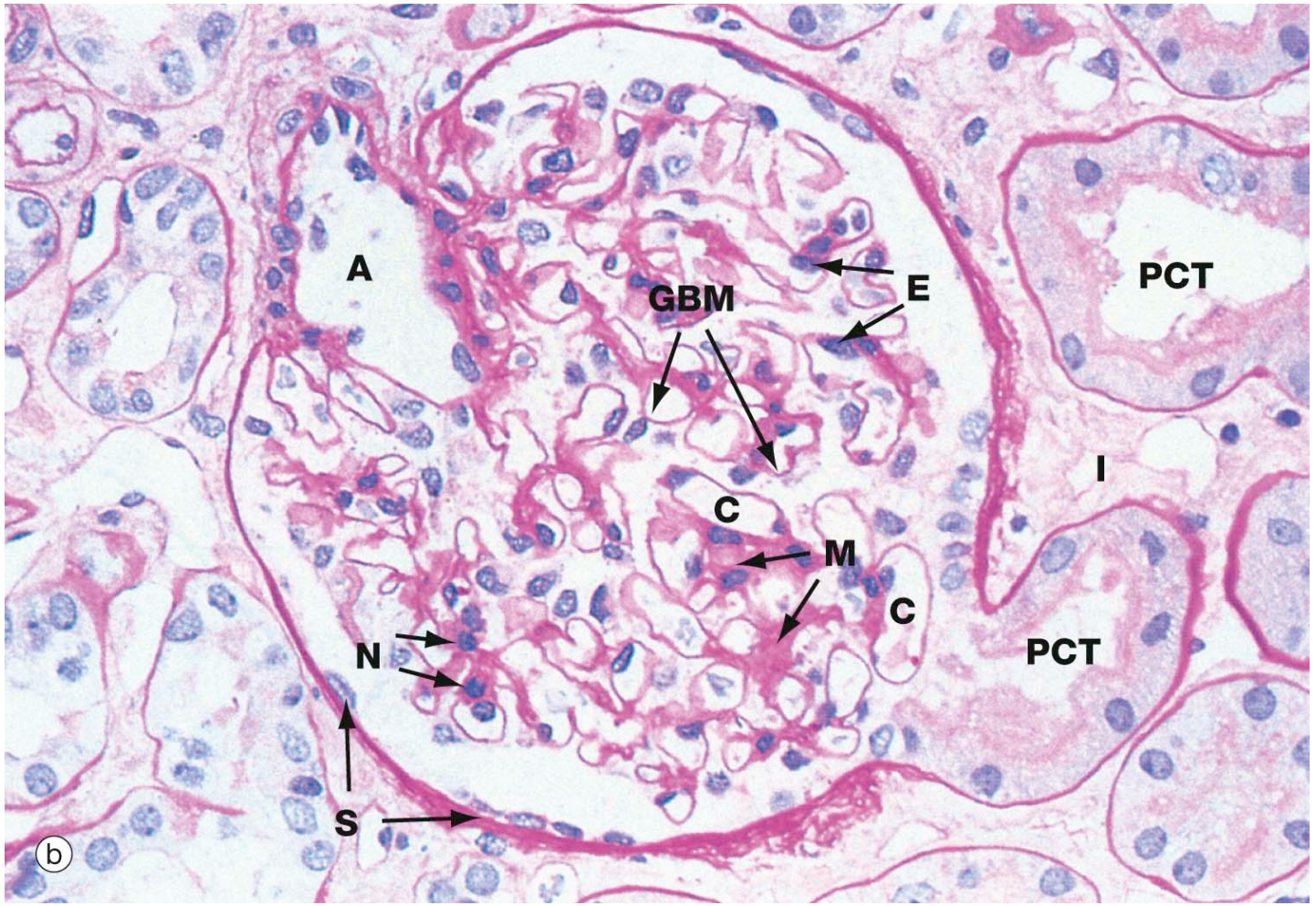
A – afferent arteriole
PCT – proximal convoluting tubules
C – glomerular capillaries
I – interstitium
M – mesangium with mesangial cell nuclei (**N**)
E – nuclei of endothelial cells
GMB – glomerular basement membrane



Renal corpuscle; PAS stain

Renal Histology

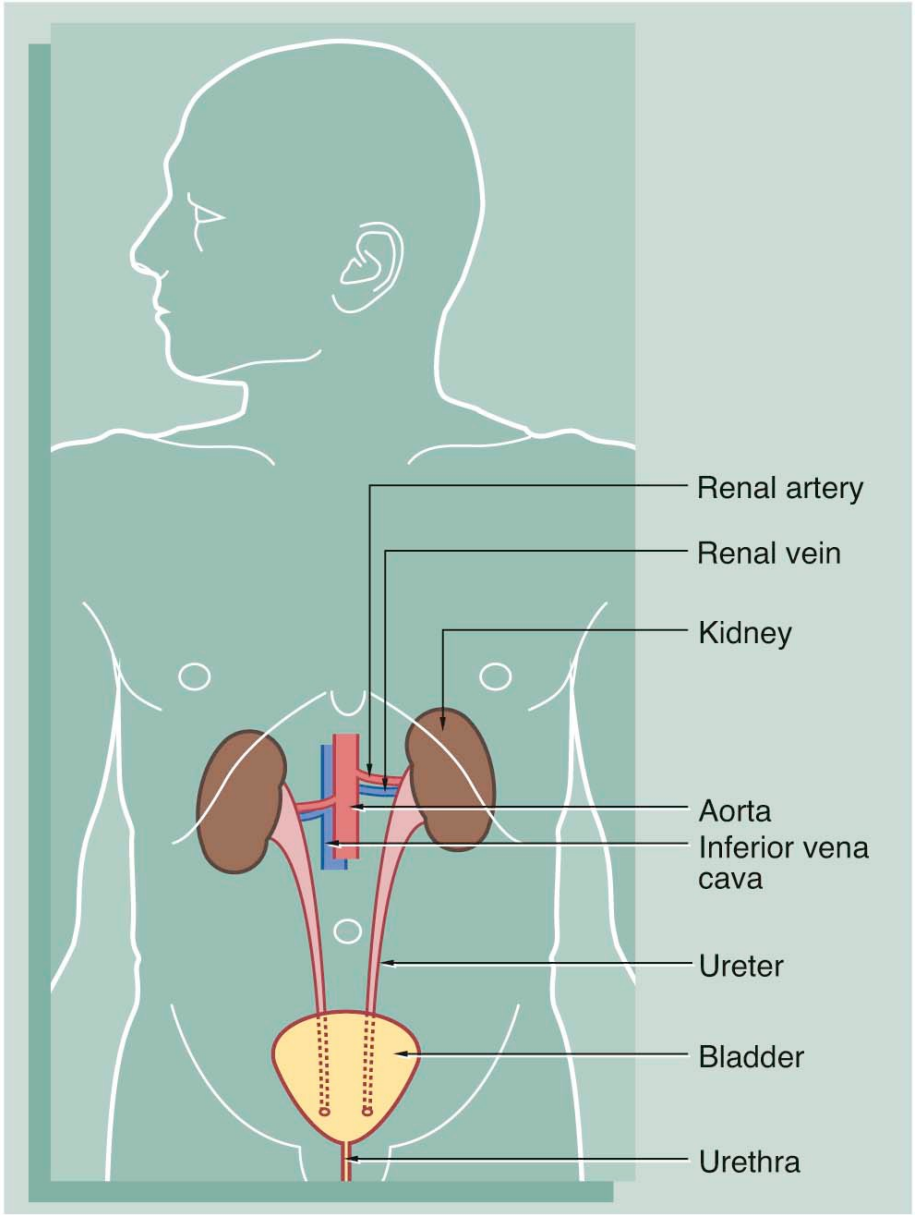
Ellis Locke, Ph.D.
Idaho College of Osteopathic Medicine

Session Objectives

1. Correlate the microscopic structure of the renal corpuscle with the production of glomerular filtrate / ultrafiltrate.
2. Describe the histological features of the proximal convoluted tubule, nephron loop, and distal convoluted tubule and correlate with the location and function of each.
3. Describe the structure of the juxtaglomerular apparatus and explain its role in the regulation of the renal and circulatory systems.
4. Distinguish the collecting duct from the nephron segments and describe its histological features and functions.
5. Relate how disruption in the cells and tissues of the kidney can lead to dysfunctions and relate these dysfunctions to common clinical correlates / patient presentations.

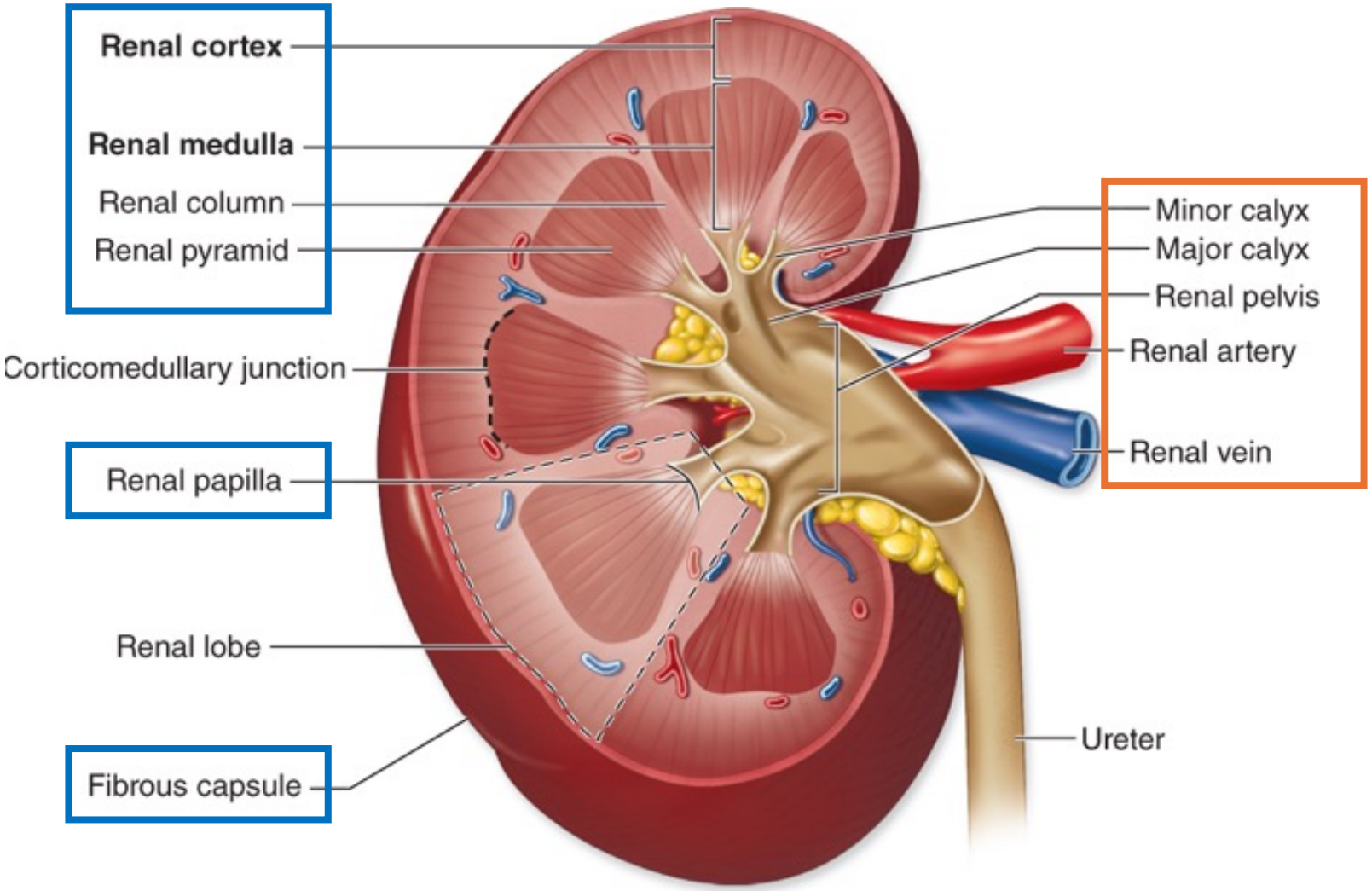
General Renal Functions

- **Maintain homeostasis:** osmoregulation and acid-base balance
- **Waste Excretion:** metabolic waste, excess water & electrolytes, and bioactive substances removed from the blood & excreted into urine
- **Endocrine functions:** synthesis of **renin** (part of RAAS control of blood pressure) and **erythropoietin** (stimulates erythrocyte production with blood O2 is low)
- **Vitamin D conversion:** hydroxylation of steroid prohormone vitamin D into active form calcitriol



Renal Gross Anatomy Review

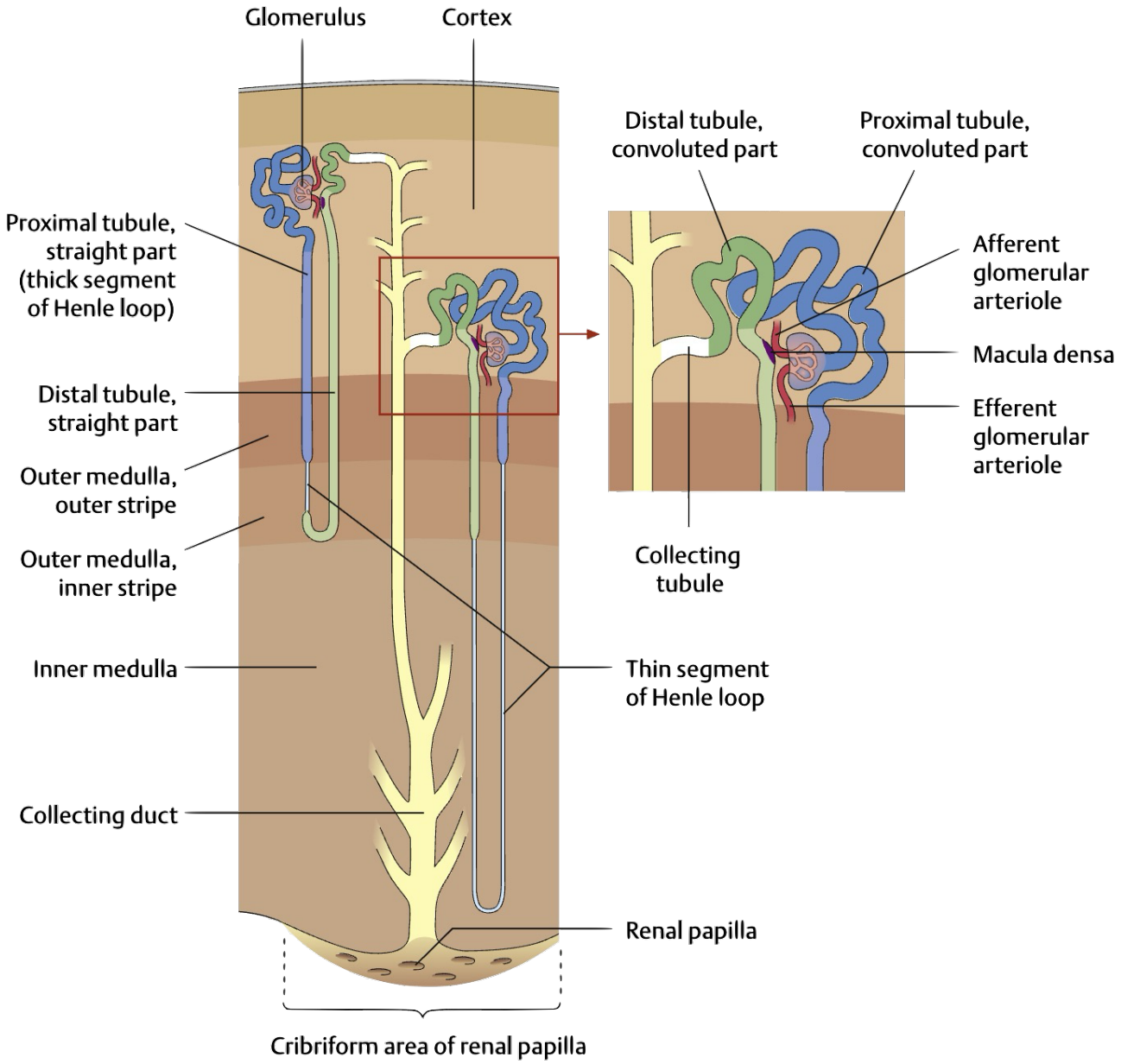
Renal Tissues



Contents of the renal sinus

Posterior view

Functional Microanatomy: Renal Cortex



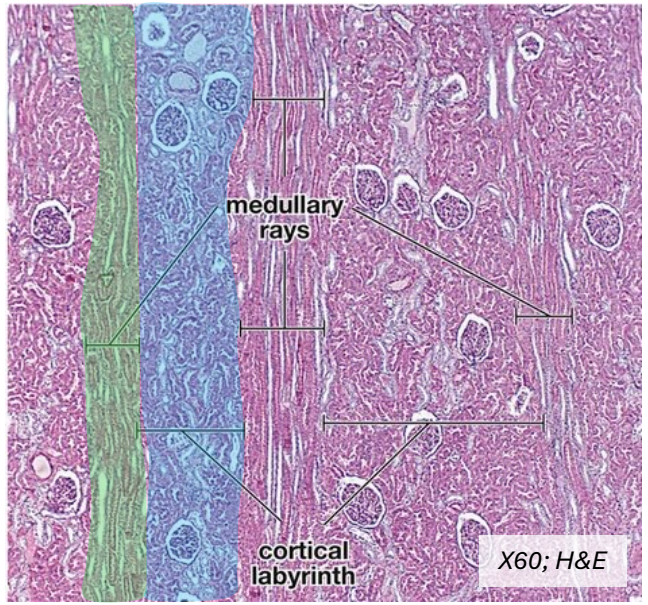
Cortical lobule

Medullary Ray

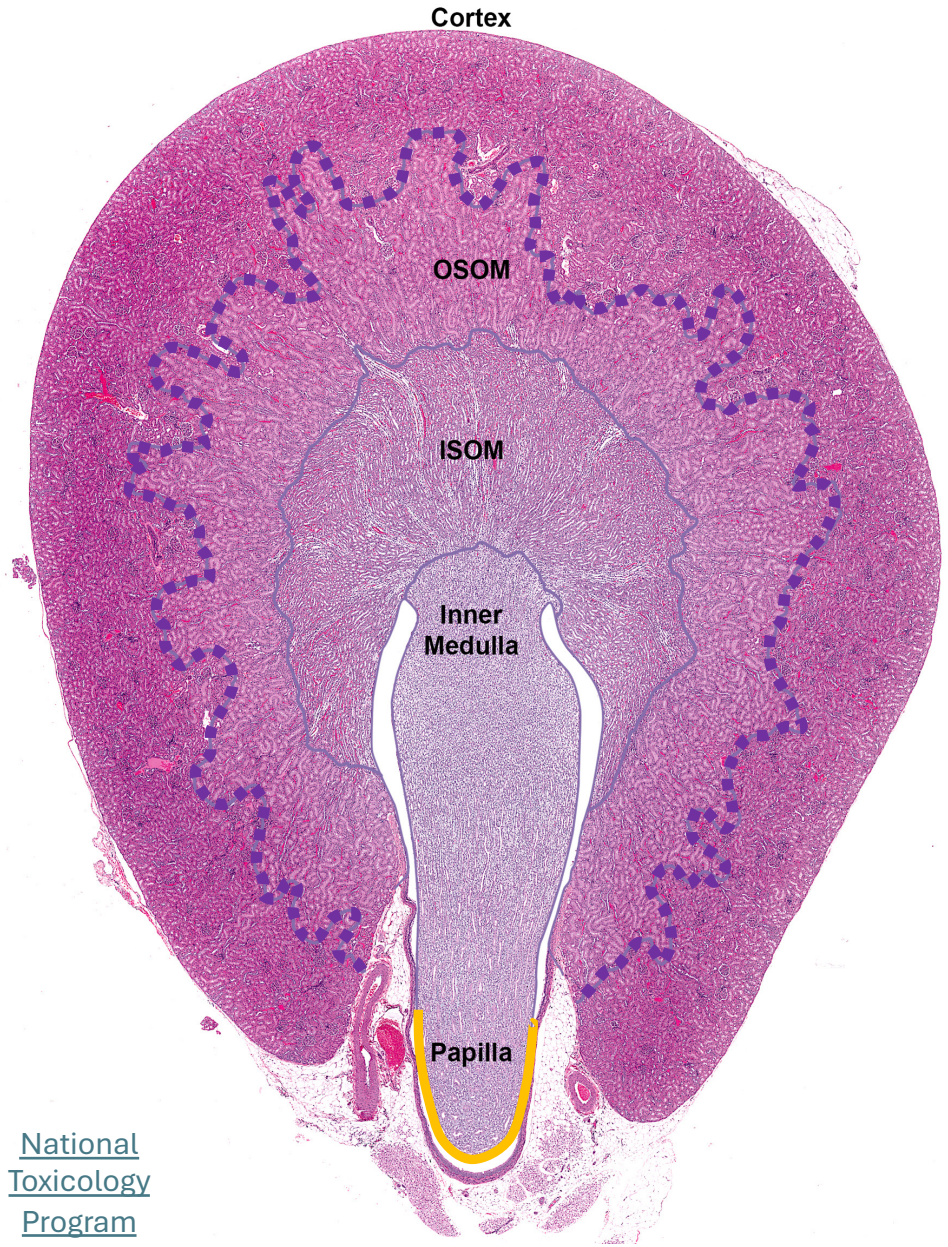
- Straight tubules (**PST** & **DST**)
- Connecting tubule & collecting duct

Cortical Labyrinth:

- Renal corpuscle
- Convoluted tubules (**PCT** & **DCT**)



Functional Microanatomy: Renal Medulla



National Toxicology Program

Medullary tissue forms **renal pyramids**
(base at ● corticomedullary junction; apex at ● papilla)

Outer Medulla

Outer Stripe of Outer Medulla (OSOM)

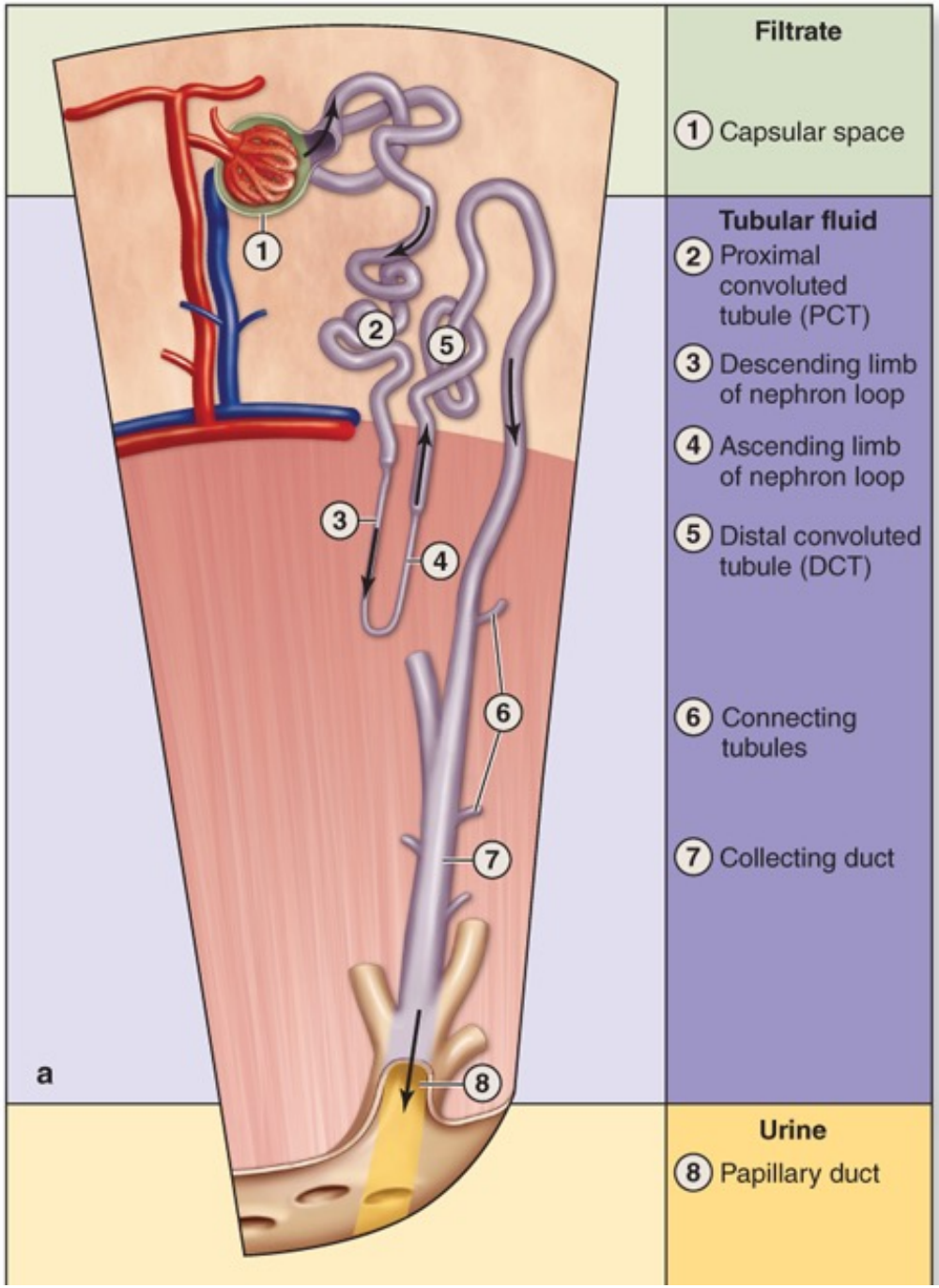
- **Thick limbs** of the nephron loop
- Continuation of the **collecting duct**

Inner Stripe of Outer Medulla (ISOM)

- **Thin descending limb** and **thick ascending limb** of the nephron loop
- Continuation of the **collecting duct**

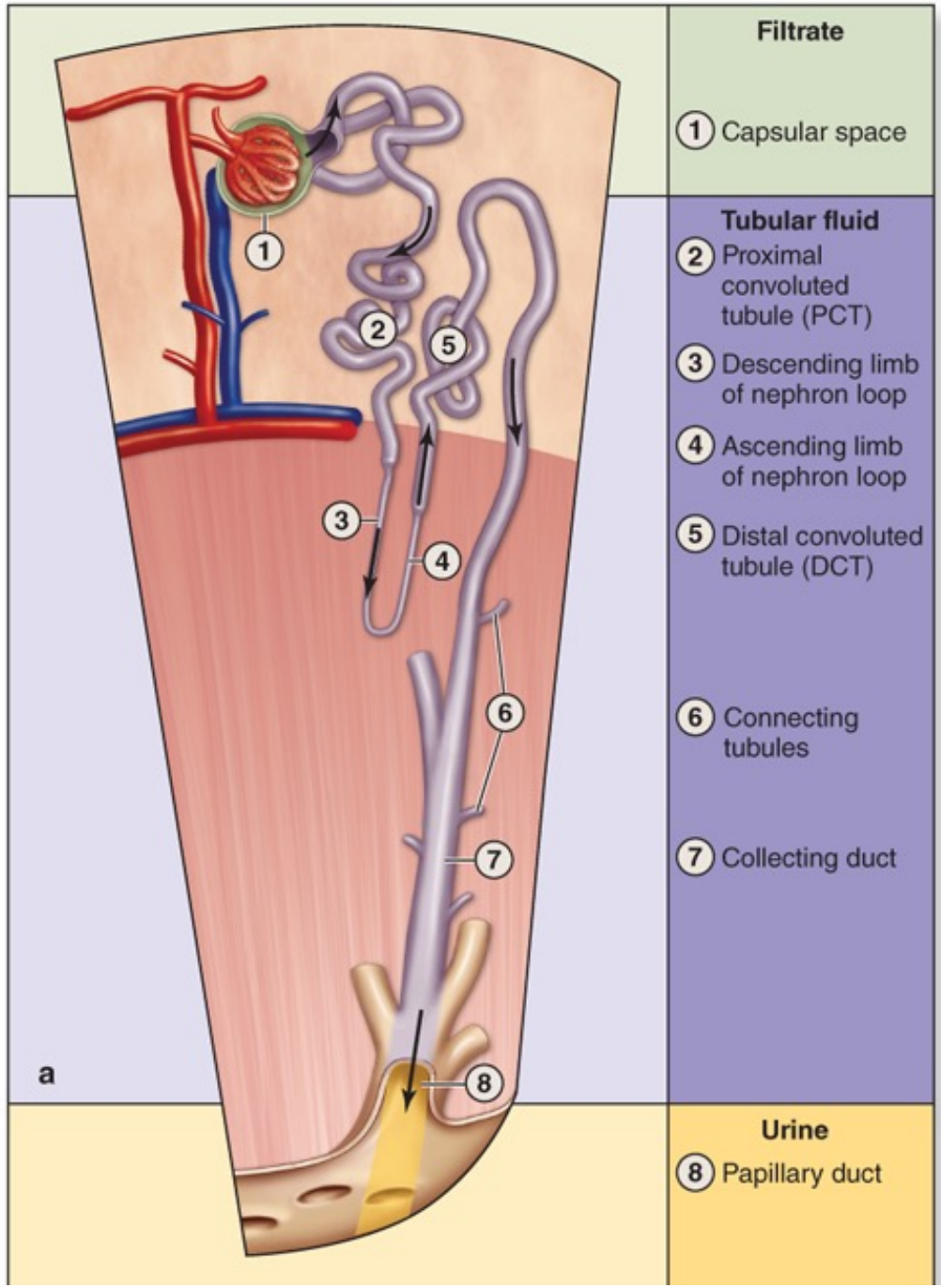
Inner Medulla

- **Thin limbs** of the nephron loop
- Continuation of the **collecting ducts** → converge to form **papillary ducts**



Uriniferous Tubule Functions

- **Glomerular filtration:** water & solutes from blood plasma exit the glomerulus and enter the lumen of the glomerular capsular space to form **ultrafiltrate**
- **Tubular reabsorption:** select substances in tubular fluid are transported across the epithelium into interstitial tissue & returned to the bloodstream
- **Tubular secretion:** substances in the interstitial tissue are transported across the epithelium & secreted into the tubular fluid
- Secretion and reabsorption mechanisms & targets (i.e., molecules being transported) vary along the length of the uriniferous tubule



Cortical vs. Medullary Tubule Functions

Cortical Functions

- **Reabsorption** to conserve nutrients, minerals, & ions needed by the body (e.g., glucose)
- **Secretion** of excess and/or toxic substances from the blood into the tubular fluid (e.g., drug clearance)

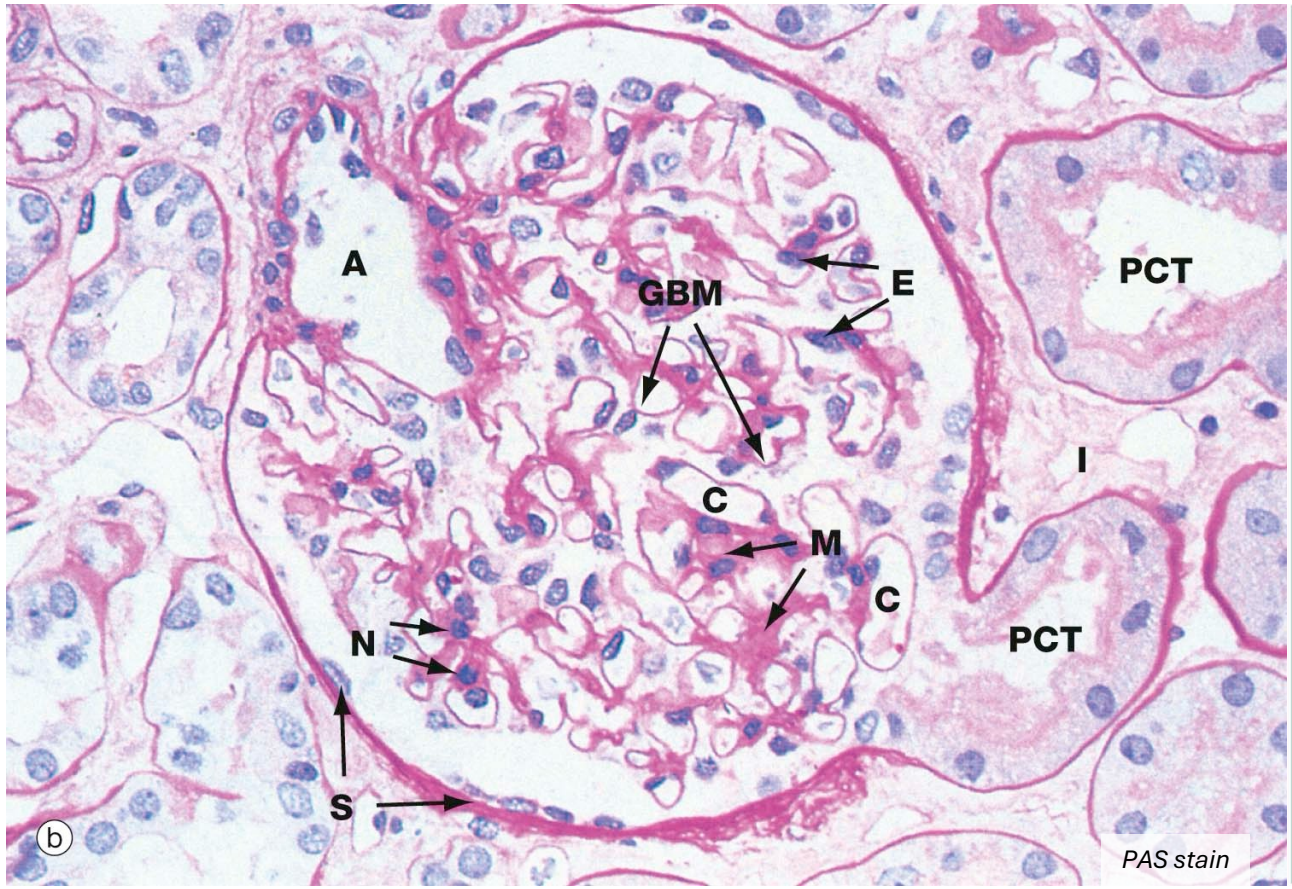
Medullary Functions

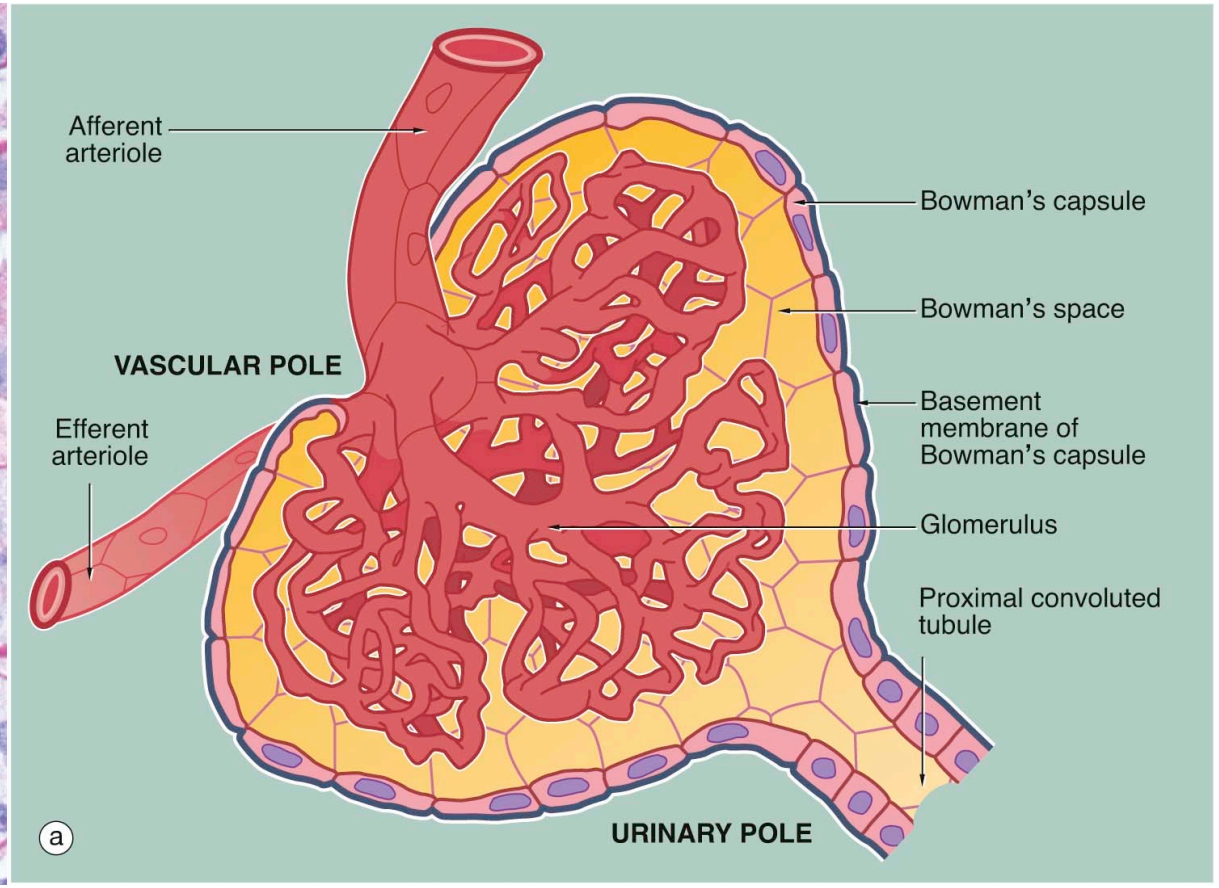
- **Reabsorption & secretion** to concentrate urea, regulate water & electrolyte balance in tubular fluid, and regulate acid-base balance

A – afferent arteriole
PCT – proximal convoluting tubules
C – glomerular capillaries
I – interstitium
M – mesangium with mesangial cell nuclei (**N**)
E – nuclei of endothelial cells
GMB – glomerular basement membrane

Renal Corpuscle

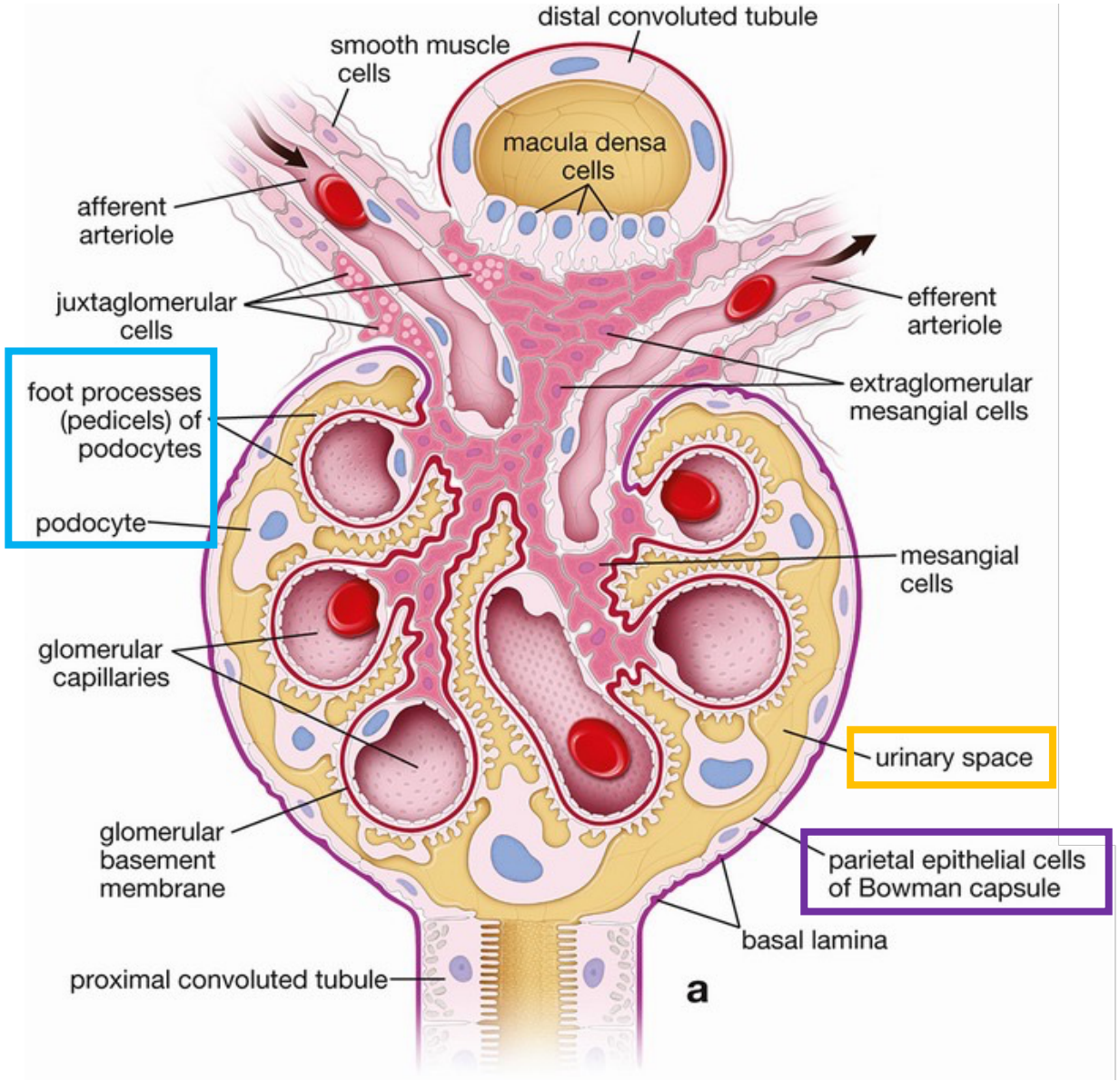
glomerulus + glomerular (Bowman's) capsule
→ site of **glomerular filtration**





Corpuscle has a **vascular pole** (efferent & afferent arterioles) and **tubular / urinary pole** (continuous with PCT)

Glomerular (Bowman) Capsule

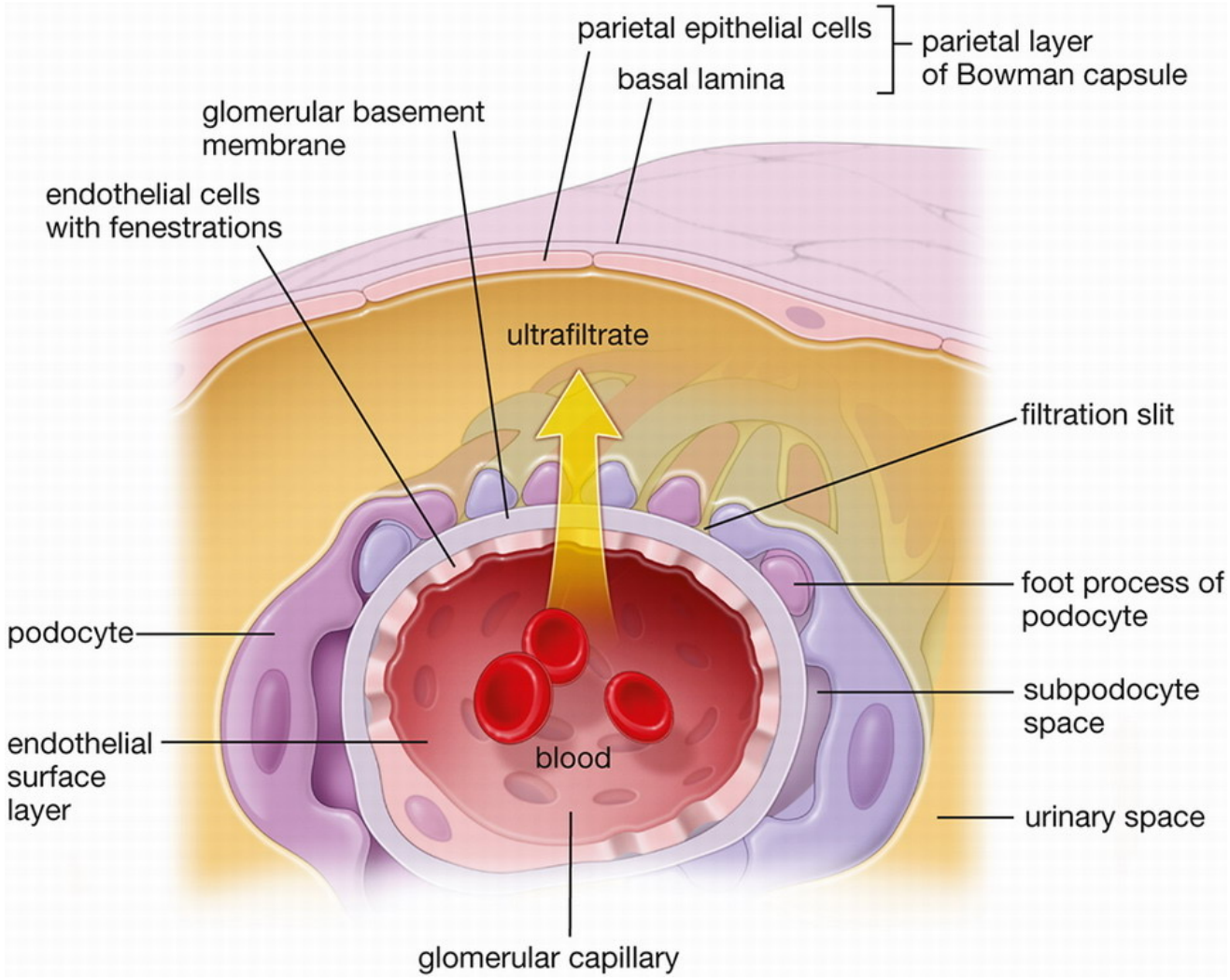


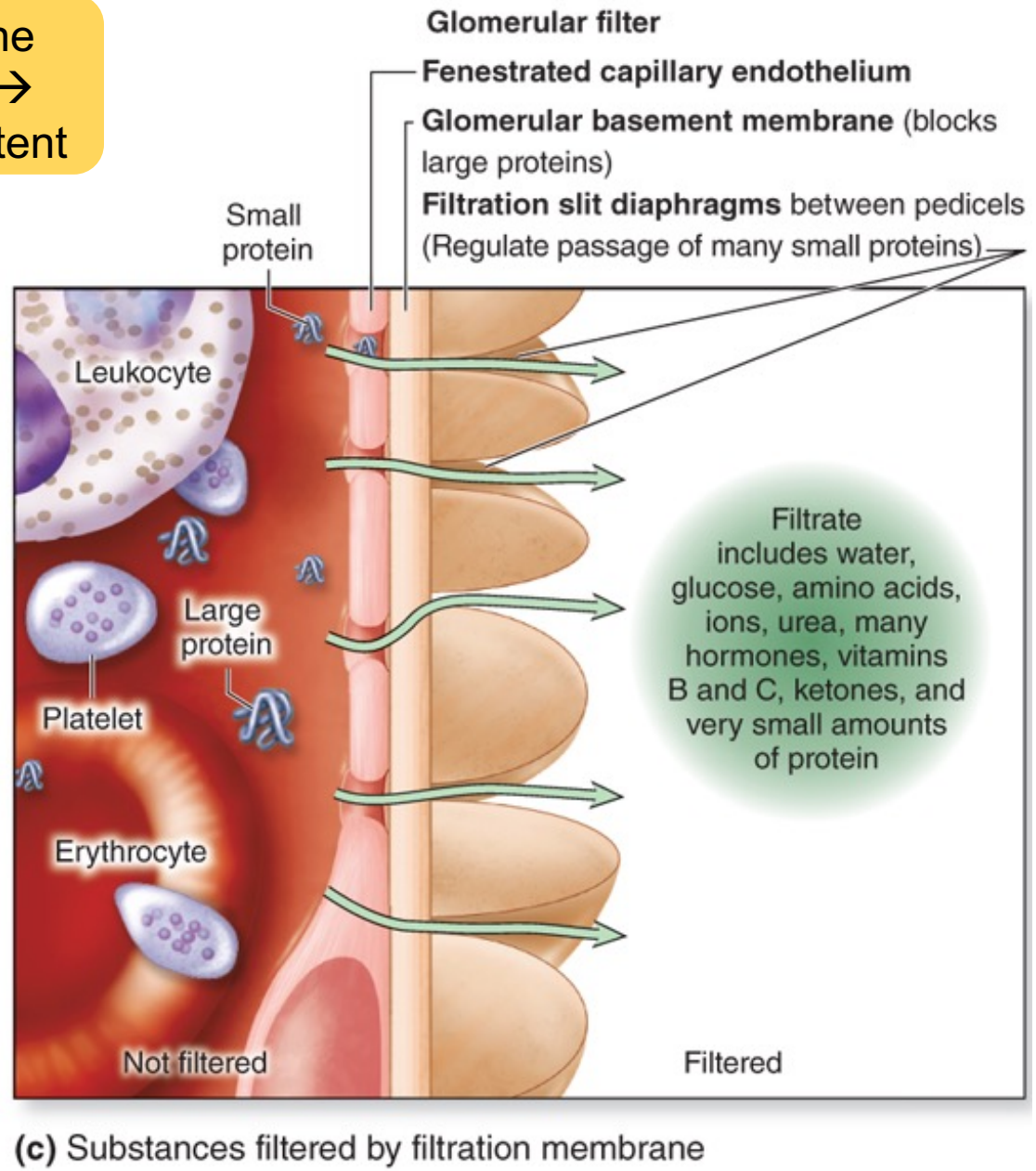
Double-layered epithelial cup surrounding the capillaries of the glomerulus

- **Parietal layer** is a simple squamous epithelium → becomes continuous with simple cuboidal epithelium of the PCT at the urinary pole of the glomerular capsule
- **Visceral layer** comprised of specialized epithelial cells called **podocytes** → contributes to the glomerular filtration barrier
- **Capsular space / urinary space** enclosed between parietal & visceral layers → receives the ultrafiltrate / glomerular filtrate

Glomerular Filtration Barrier

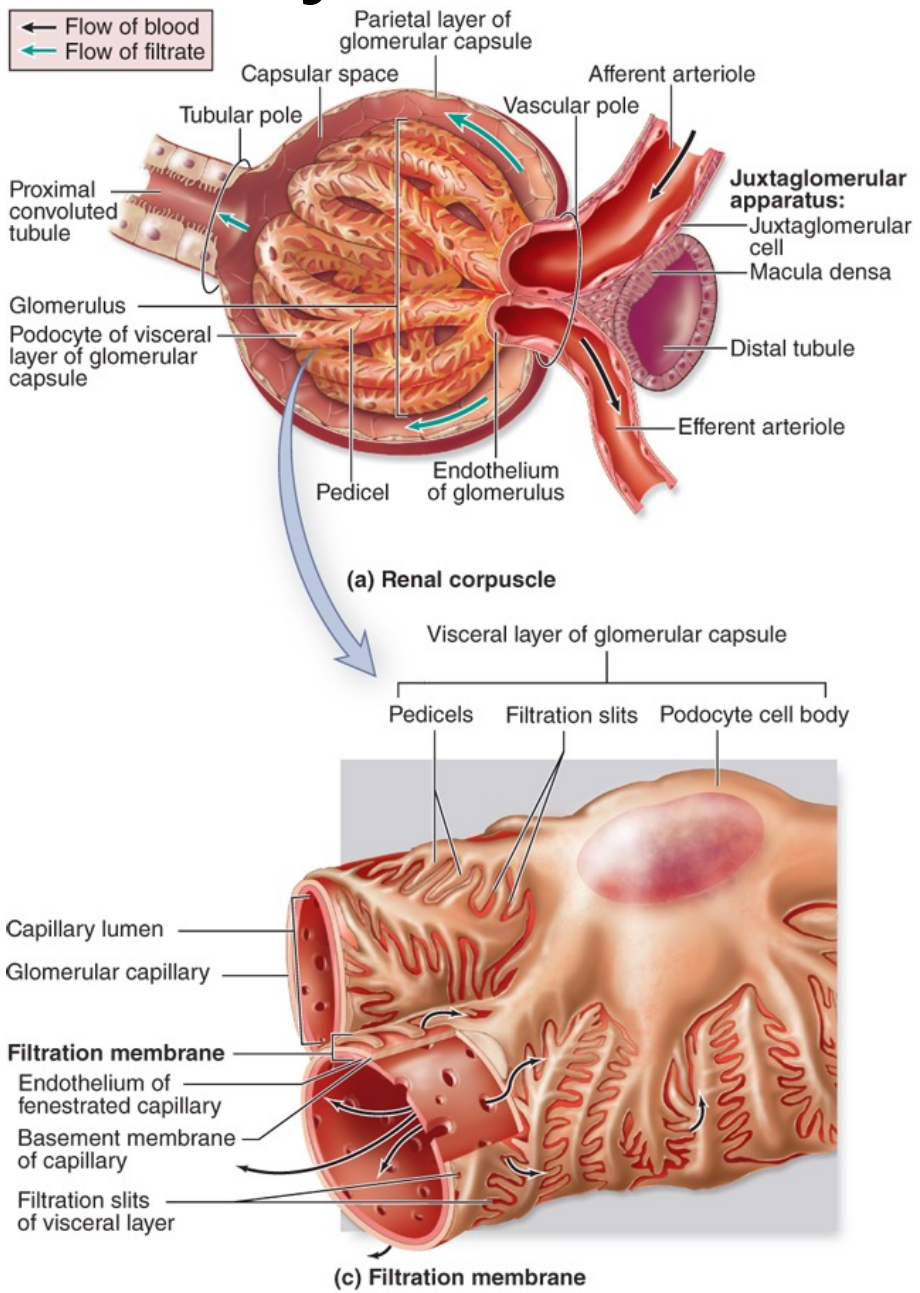
~20% of plasma entering the glomerulus filters into the capsular space as **glomerular filtrate / ultrafiltrate** → similar composition to plasma but very low protein content





(c) Substances filtered by filtration membrane

Layers of the Glomerular Filtration Barrier



Fenestrated capillary endothelium

- Fenestrations larger, more numerous, and more irregular than fenestrations in other capillaries
- Allows passage of plasma fluid, solutes, & some small proteins
- Restricts passage of blood cells and platelets

Glomerular basement membrane

- Fused basement membranes of the fenestrated capillary endothelium & podocyte epithelial cells of the visceral capsule
- Major functional component of the GFB – acts as both a physical barrier and an ion-selective filter (negatively charged)
- Restricts passage of large plasma proteins & organic anions

Podocytes

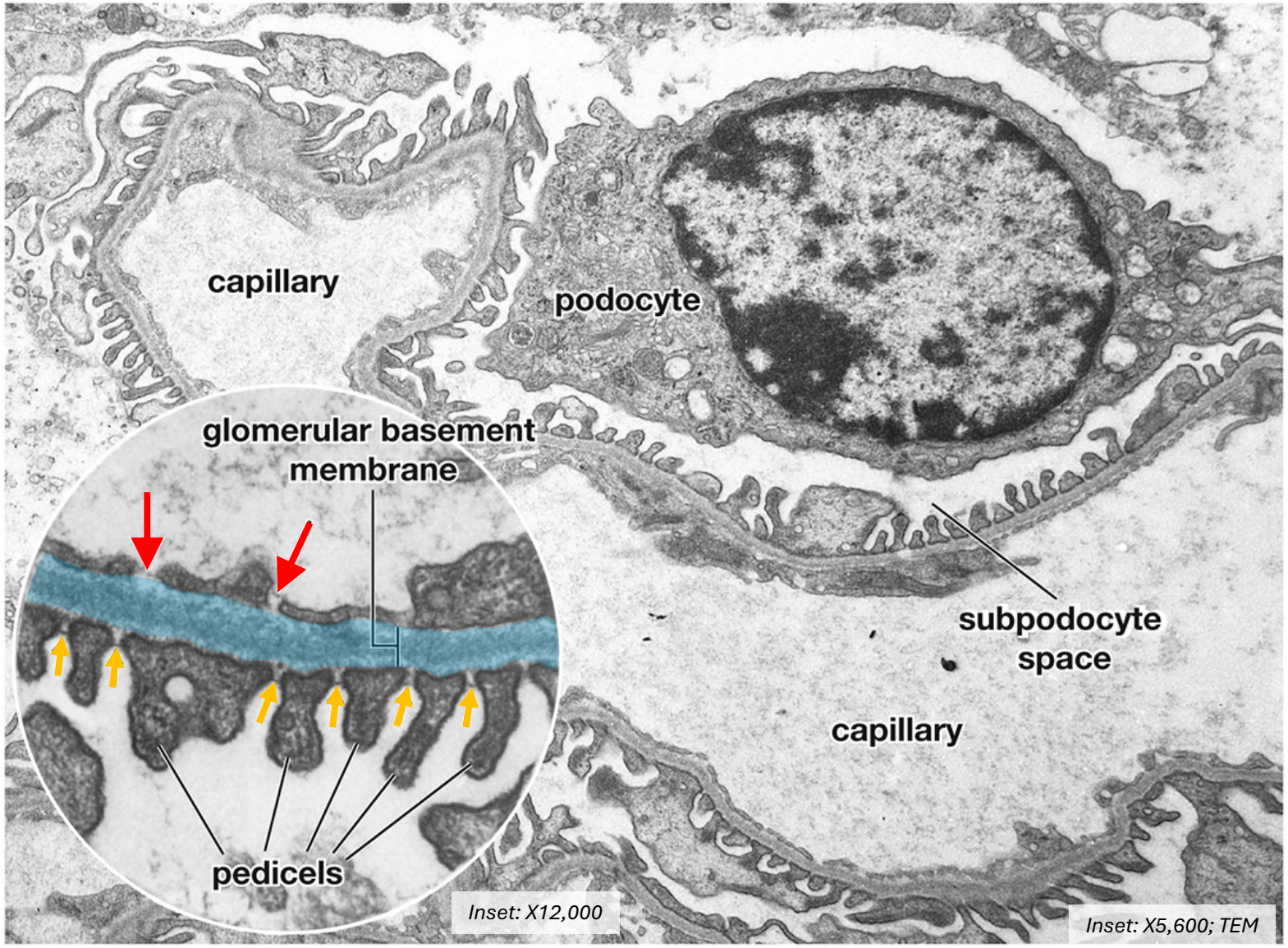
- **Stellate epithelial cells** of the visceral capsule layer
- Multiple primary processes with numerous secondary processes (**pedicels** / **foot processes**) that interdigitate to form **“filtration slits”**
- Size-selective filter that blocks some small proteins & organic anions

Layers of the Glomerular Filtration Barrier

Endothelial fenestrations
in the glomerular capillary

Glomerular basement membrane
(fused BM of capillary endothelium & podocytes)

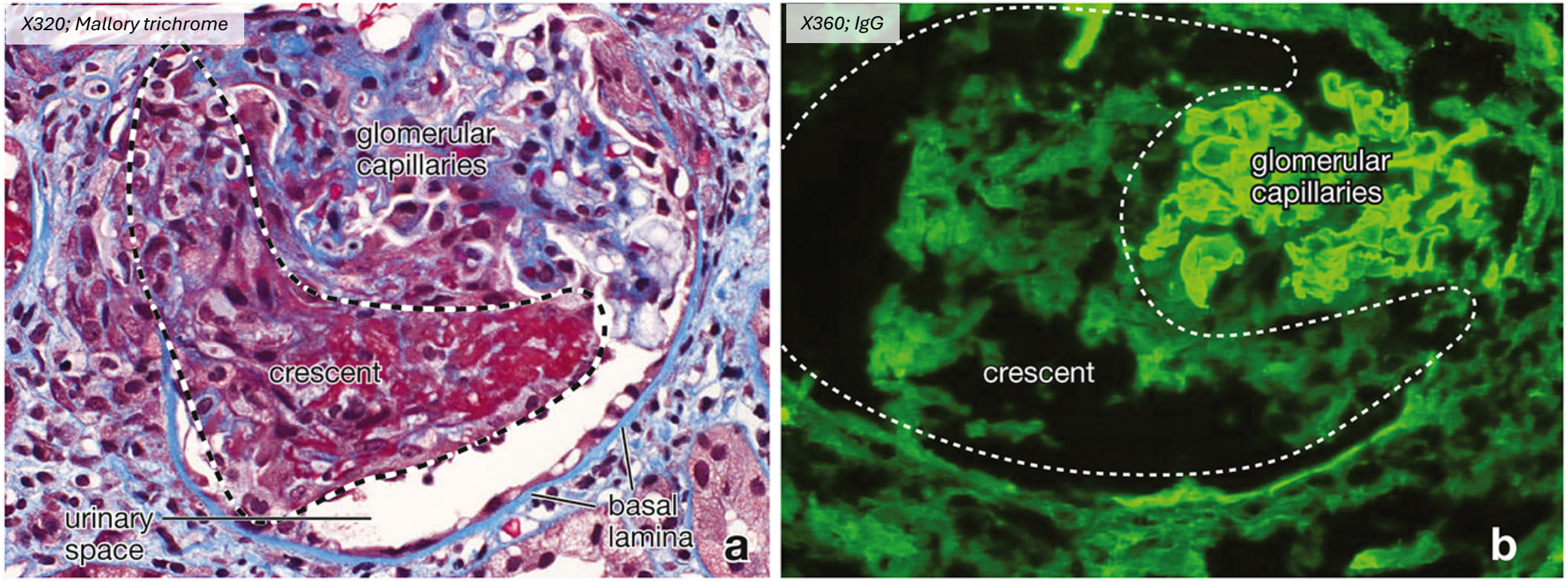
Filtration slit diaphragms
between foot processes (pedicels) of podocytes



Clinical Correlate: Goodpasture Syndrome

Anti-GBM antibody-induced glomerulonephritis

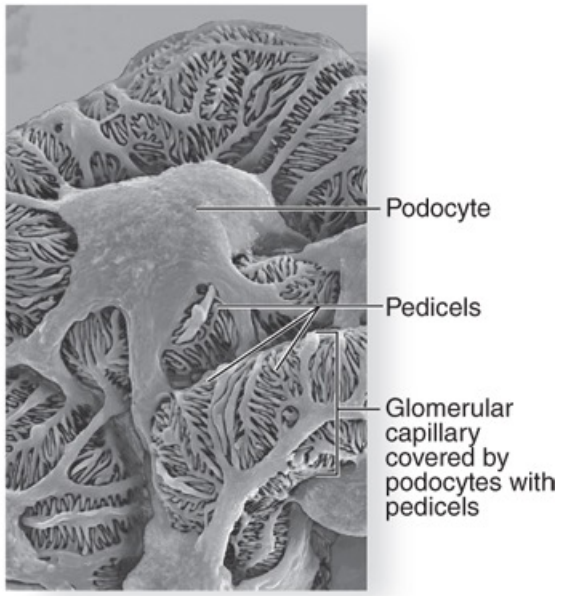
Autoimmune response to type IV collagen molecules in the GBM → progressive glomerulonephritis & pulmonary hemorrhage (disrupted blood-air barrier in the alveoli)



Crescent-shaped glomeruli characteristic of glomerulonephritis

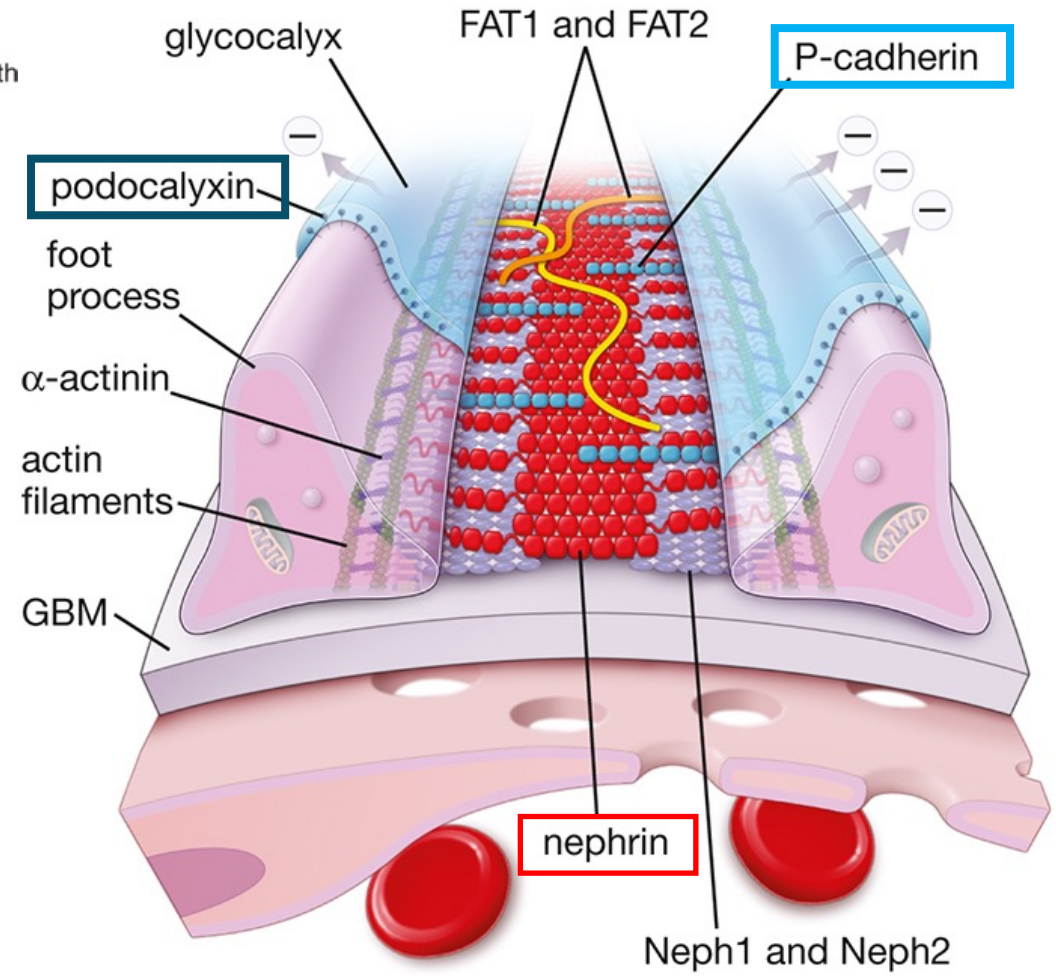
UpToDate

Filtration Slit Diaphragms



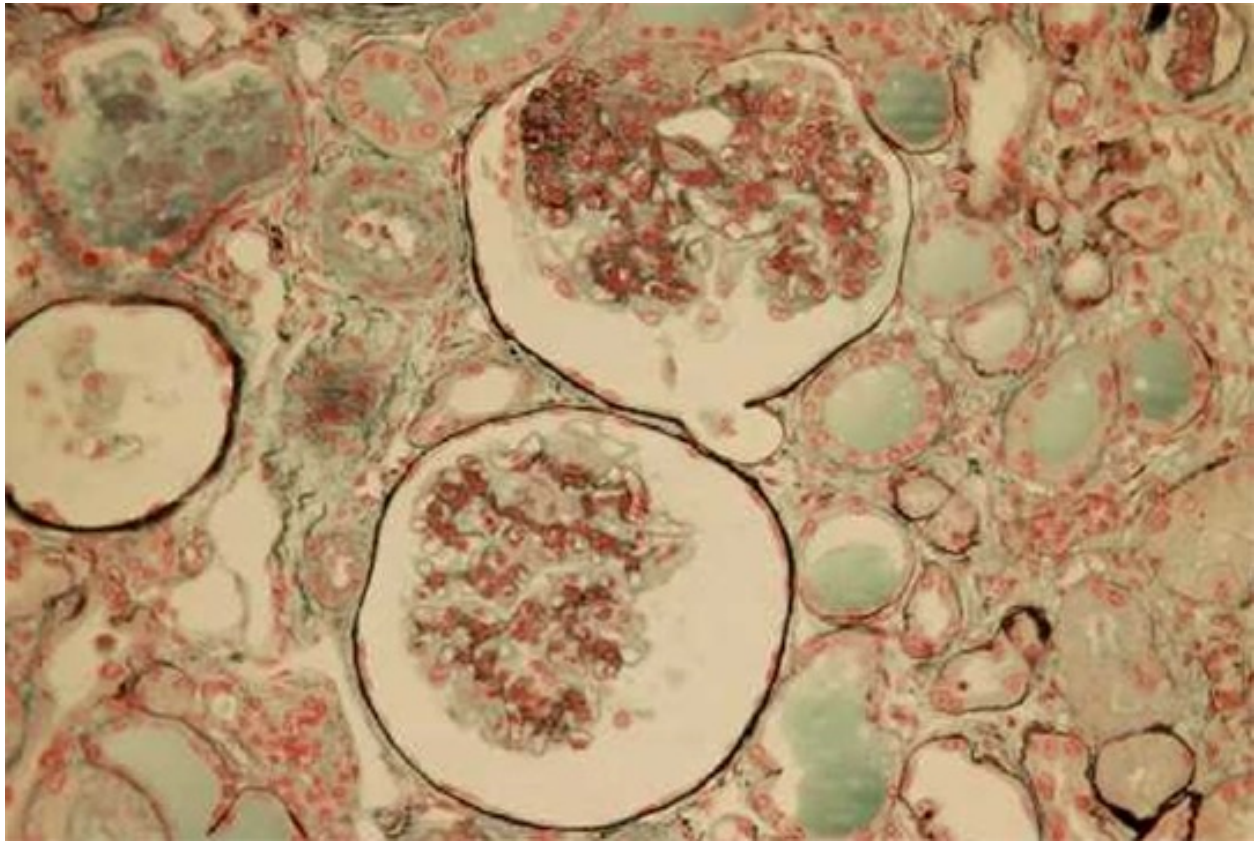
(d) Podocytes

Filtration slits between interdigitating podocyte foot processes are bridged by an ultra-thin, specialized **filtration slit diaphragm**



- Occluding (tight) junctions formed by **nephrin** and **P-cadherin**
- Glycocalyx containing negatively-charged **podocalyxin** extends up foot processes to keep slit open
- Forms a negatively charged molecular mesh that repels organic anions & some polypeptides in the capillary

Clinical Correlate: Congenital Nephrotic Syndrome



"Kidney biopsy from an infant with congenital nephrotic syndrome of the Finnish type… Histologic changes include the characteristic findings of mild mesangial hypercellularity and increased mesangial matrix in the glomeruli, as well as irregular microcystic dilatation of proximal tubules." – UpToDate

Typically result of single gene mutations

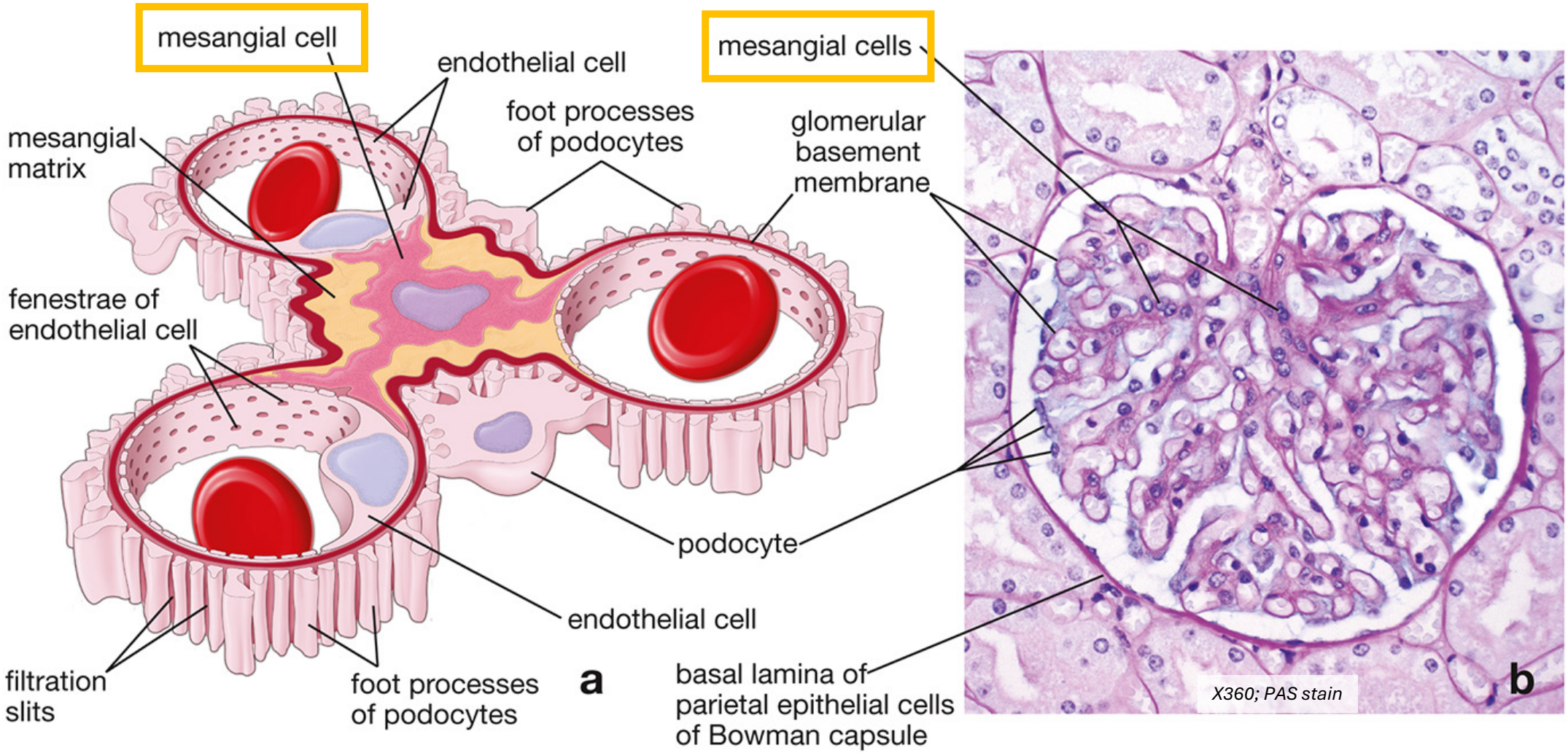
Congenital nephrotic syndrome of the Finnish Type (CNF) is most common & is due to autosomal recessive mutations in the **NPHS1 gene** → encodes for the **nephrin protein** in the filtration slit diaphragm

Clinical presentation at birth includes **edema** and **proteinuria**

Renal failure typically occurs between 3 – 8 years of age

Mesangial Cells

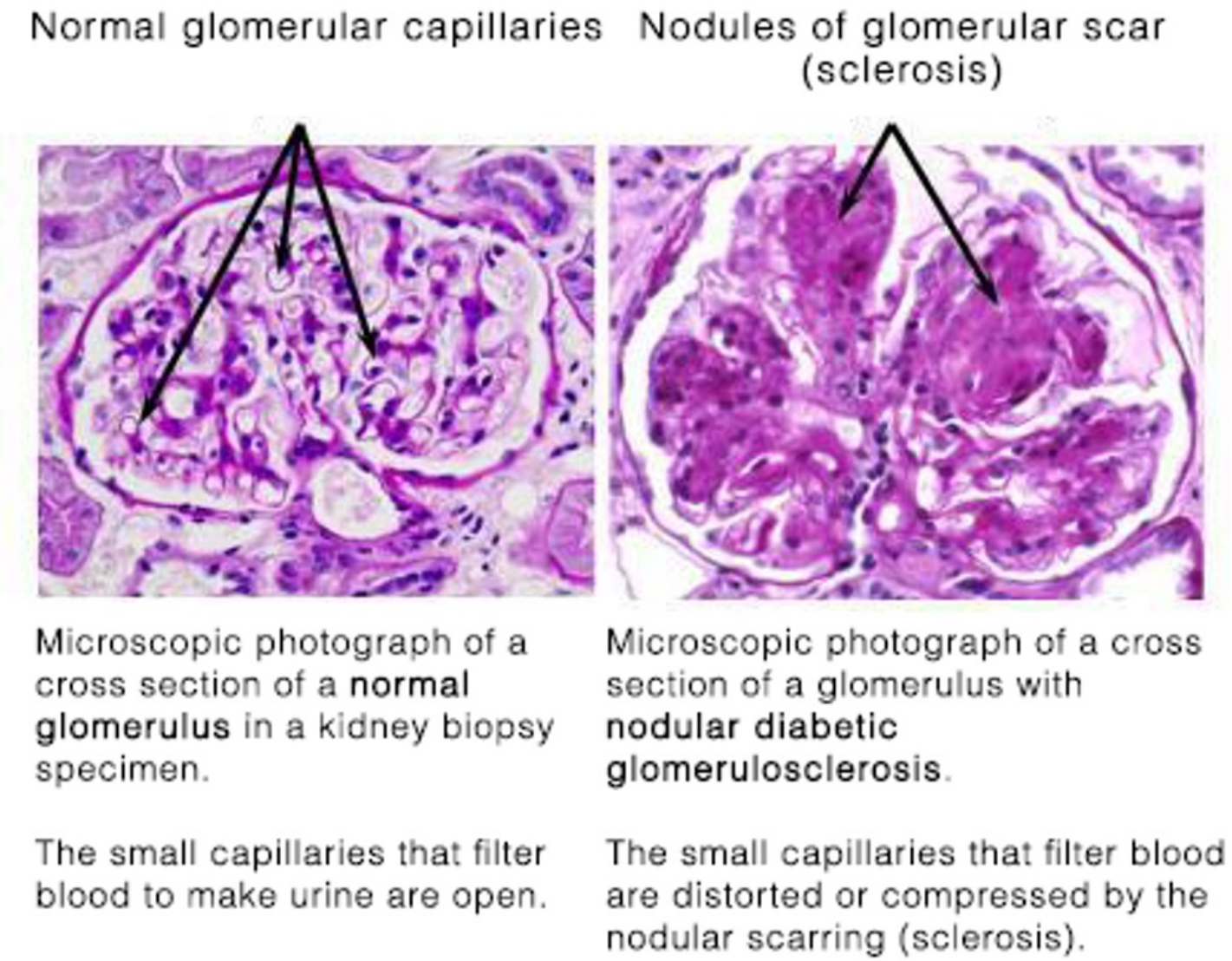
Functions include **phagocytosis** of proteins adhered to the GFB, **structural support** for capillaries, **secretion** of factors involved in immune defense & glomerular repair, and **contractile properties** adjust capillaries in response to local blood pressure changes



Clinical Correlate: Diabetic Glomerulosclerosis

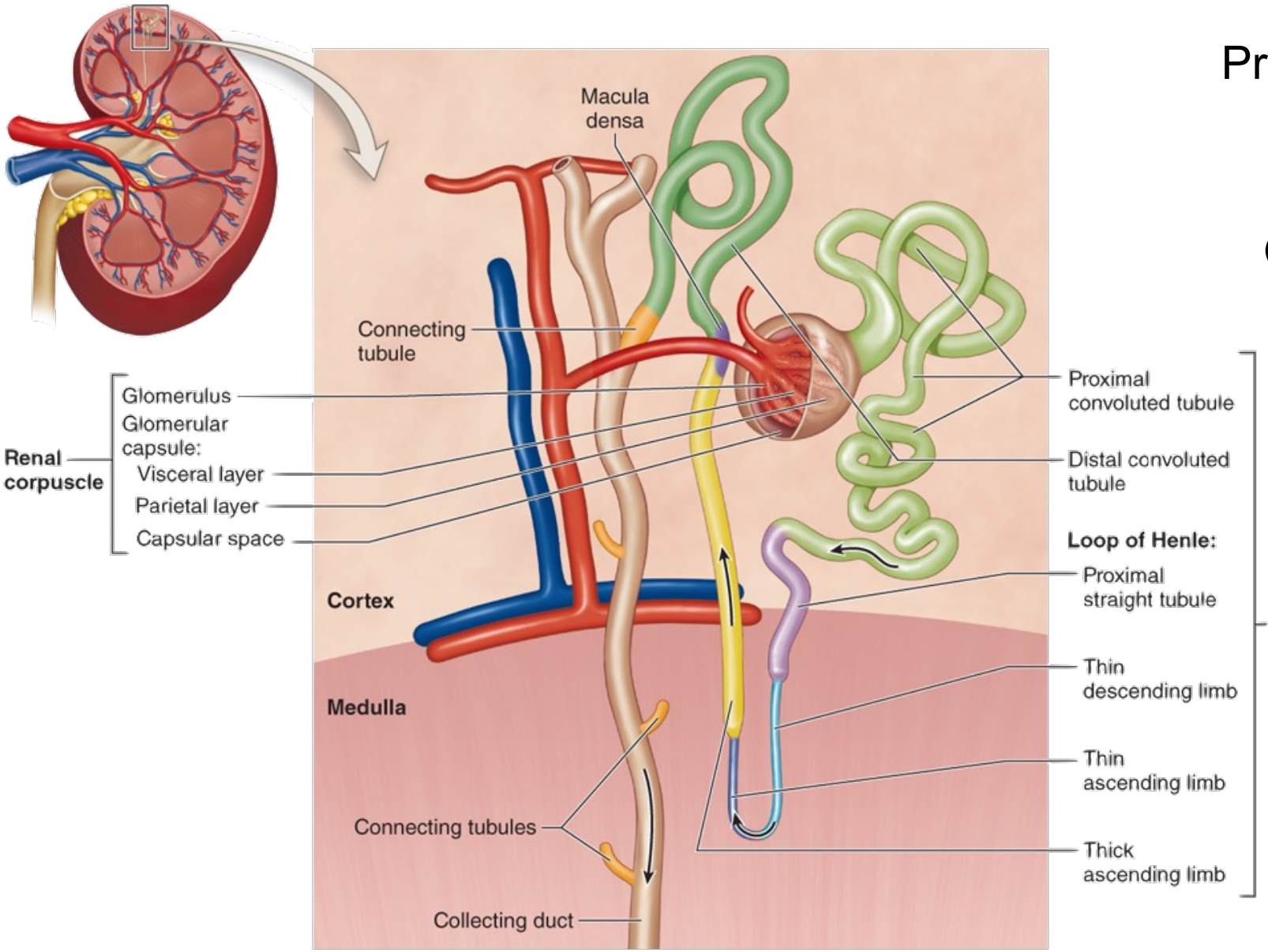
Thickening of the GBM due to systemic microvascular sclerosis

Leading cause of end-stage kidney disease in the United States



UNC Kidney Center

Review: Nephron Tubule



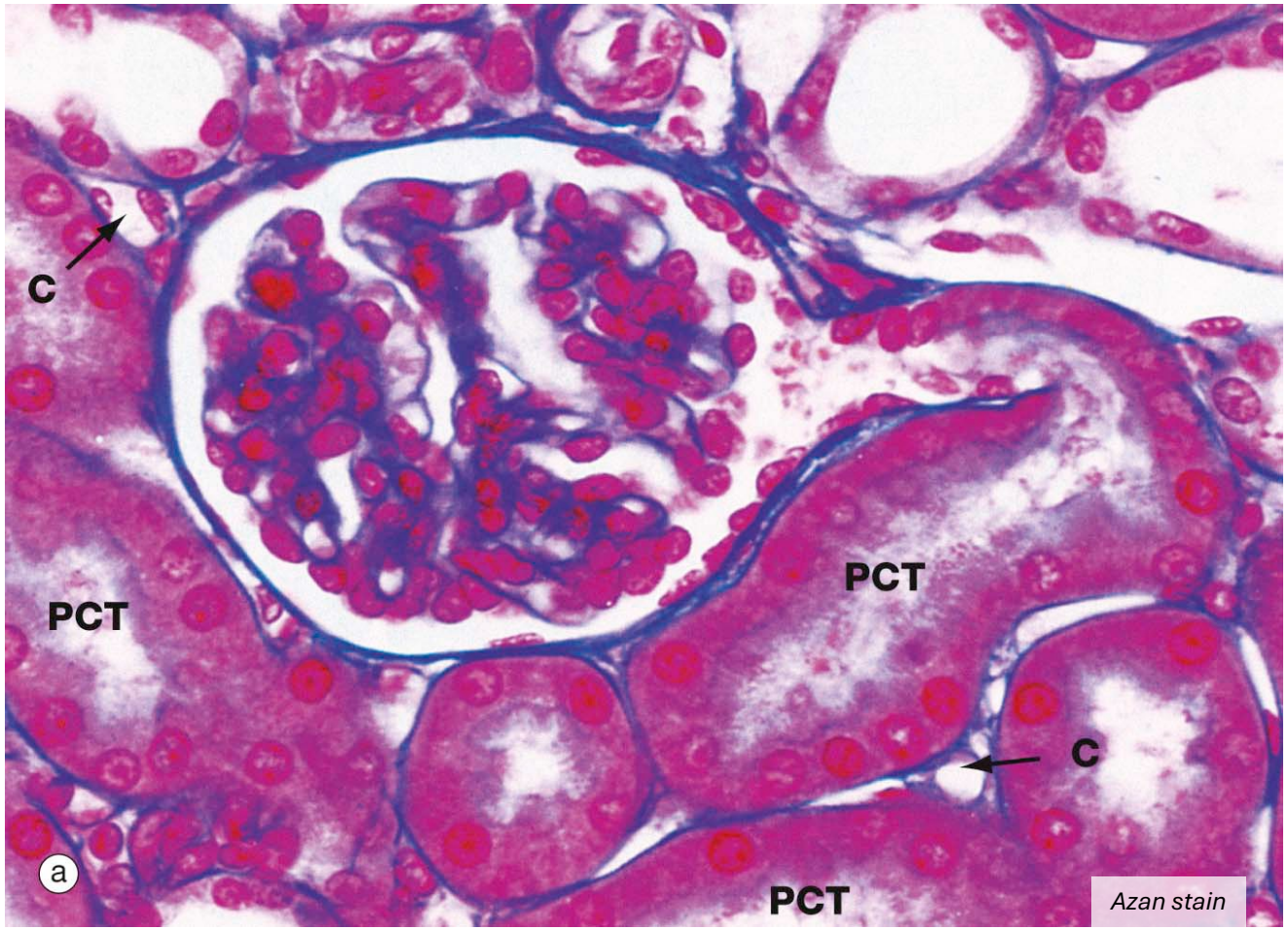
Process fluid before it enters the collecting system

- **Proximal convoluted tubule**
- **Nephron loop** (of Henle)
  - **Thick descending limb** (proximal straight tubule)
  - **Thin descending limb**
  - **Thin ascending limb**
  - **Thick ascending limb** (distal straight tubule)
- **Distal convoluted tubule**

Proximal Convoluted Tubule

BB – brush border
BM – basement membrane
C – peritubular capillary
PCT – proximal convoluted tubule

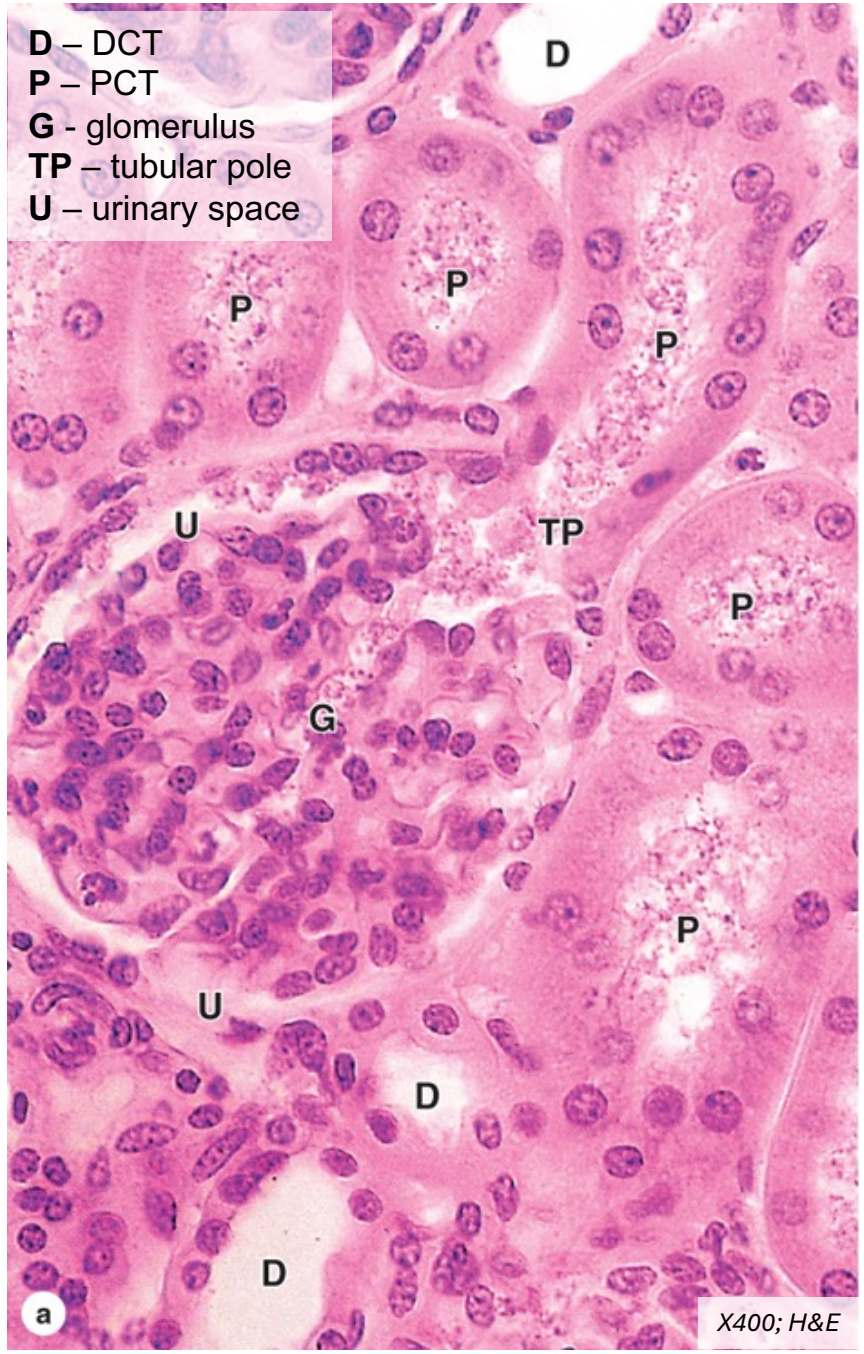
Initial major site of **reabsorption**



Key functions:

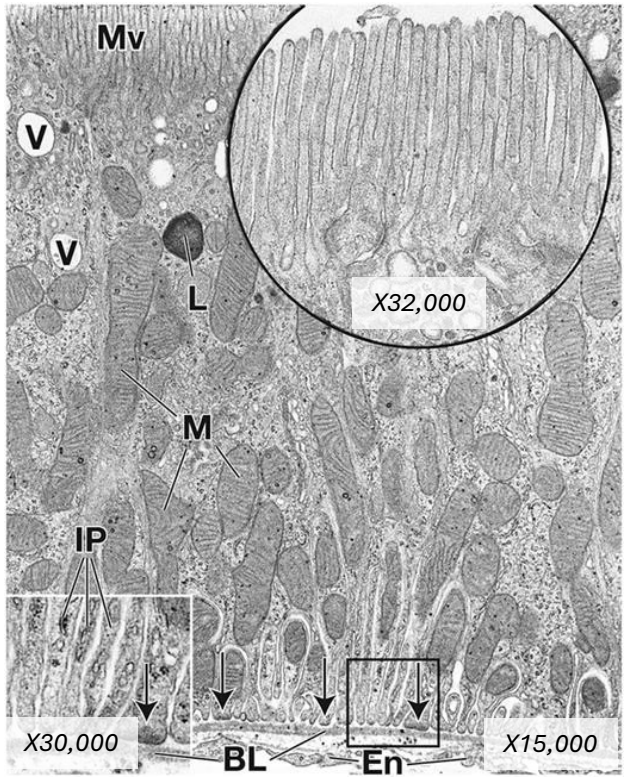
- Reabsorption of nearly all organic nutrients (e.g., **glucose**, amino acids)
- Reabsorption of **Na^+** (active transport) and **Cl^-** (passive transport)
- Reabsorption of of **H_2O** (aquaporin channels & leaky tight junctions)
- Secretion of molecules picked up from surrounding peritubular capillaries → mechanism of drug clearance
- Hydroxylation of vitamin D

Proximal Convoluted Tubule



Key histological features:

- Simple cuboidal epithelium with **prominent brush border** of densely packed **microvilli** → lumen appears irregular
- Boundaries between cells not visible due extensive **lateral interdigitations**; large cell size → not all nuclei visible in routine section



- Typically stain more acidophilic than DCT epithelial cells
- Basal domain contains elongated, longitudinally-oriented mitochondria

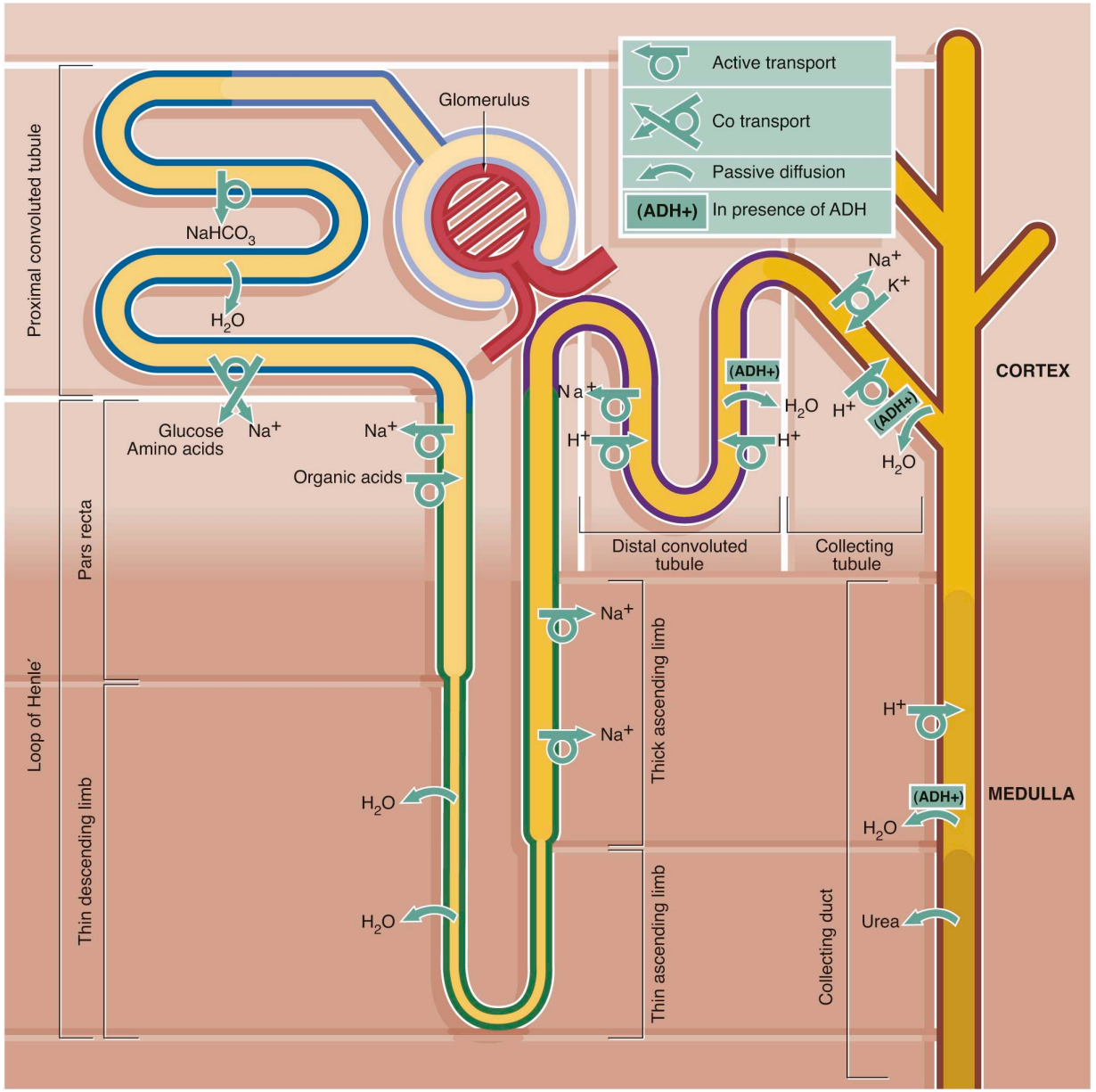
V – vesicles
L – lysosomes
MV – microvilli
M – mitochondria
BL – basal lamina
IP – interdigitating processes
En – Endothelium of peritubular capillary

Nephron Loop

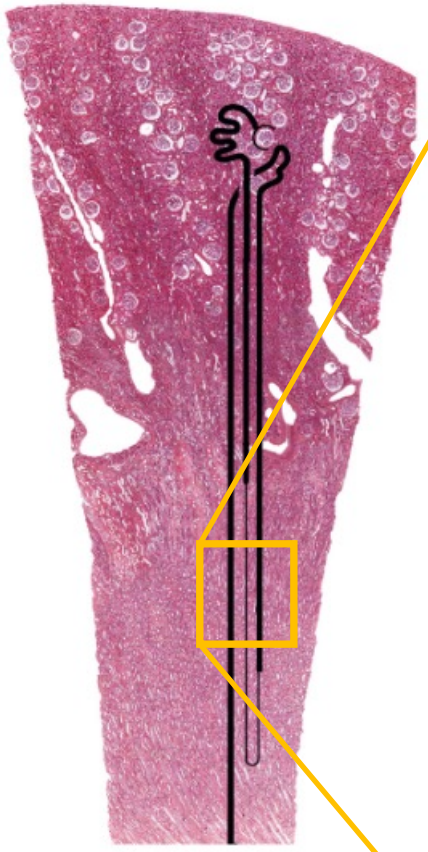
Creates & maintains **osmotic gradient** in the medulla that allows reabsorption of water & ions → **concentration of fluid**

Key functions:

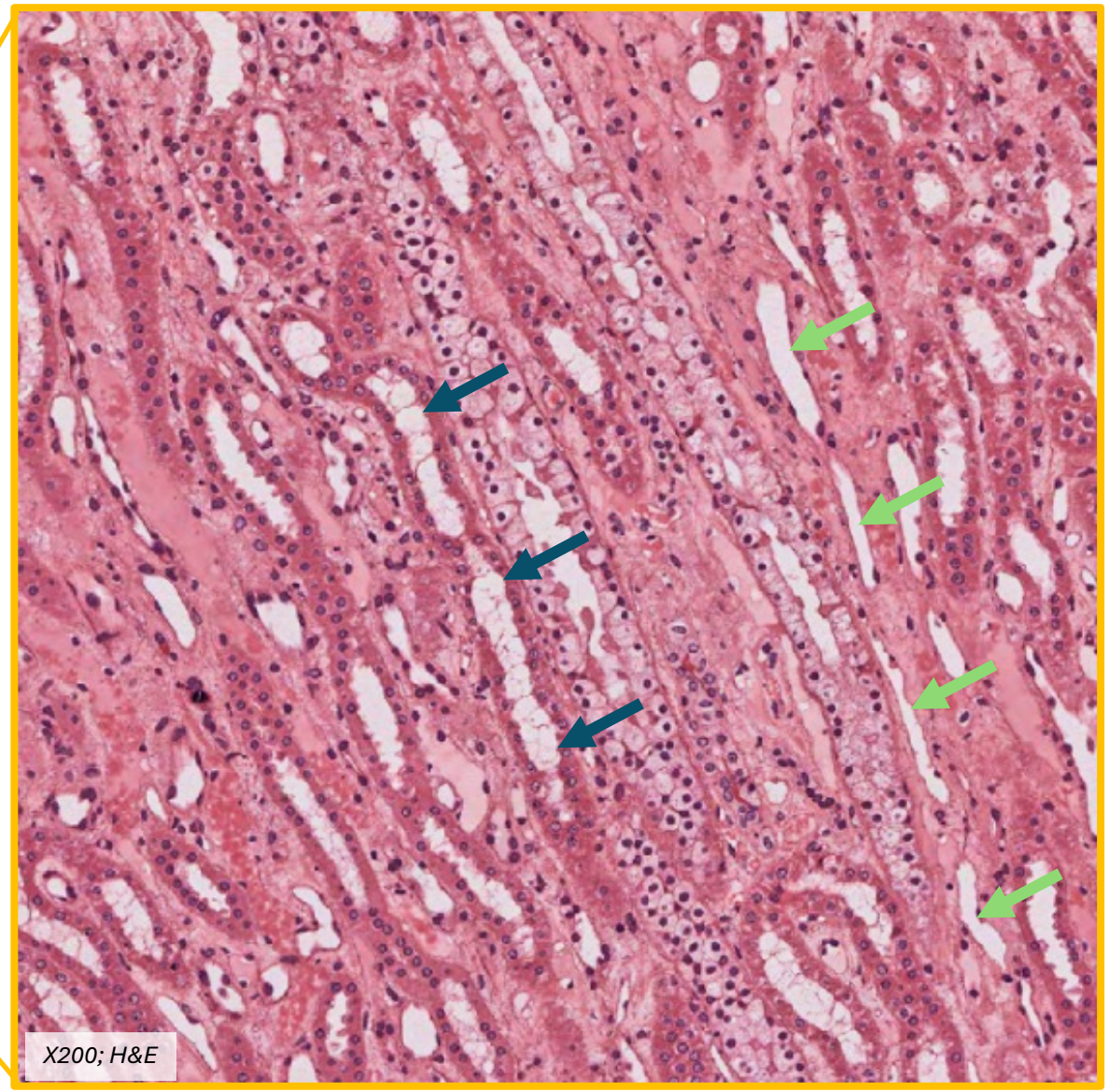
- **Thick descending limb (PST):** reabsorption of remaining glucose from tubular fluid
- **Thin descending limb:** highly **permeable to H_2O** (aquaporin channels), permits only small amount of Na+ & urea to enter lumen
- **Thin ascending limb:** highly **permeable to Na^+, Cl^-, and K^+** (active & passive transport) but **impermeable to H_2O**
- **Thick ascending limb (DST):** specialized for active transport of Na+



Nephron Loop



X20; H&E



X200; H&E

Key histological features:

Thick limbs

Simple cuboidal epithelium

- Thick descending / PST: continuous with PCT, retains **brush border**
- Thick ascending / DST: continuous with DCT, no brush border (uniform lumen)

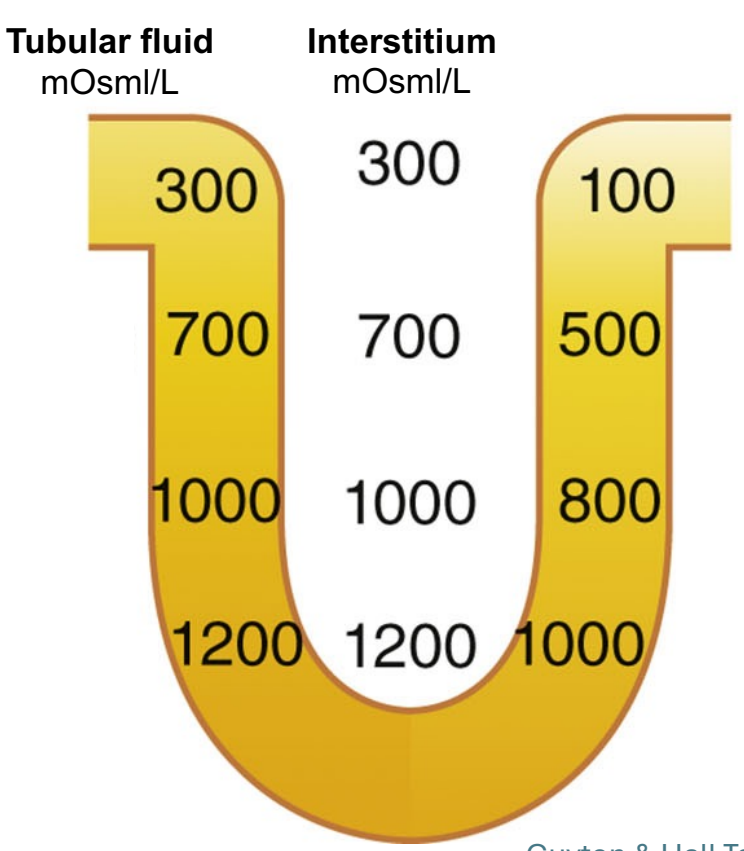
Thin limbs

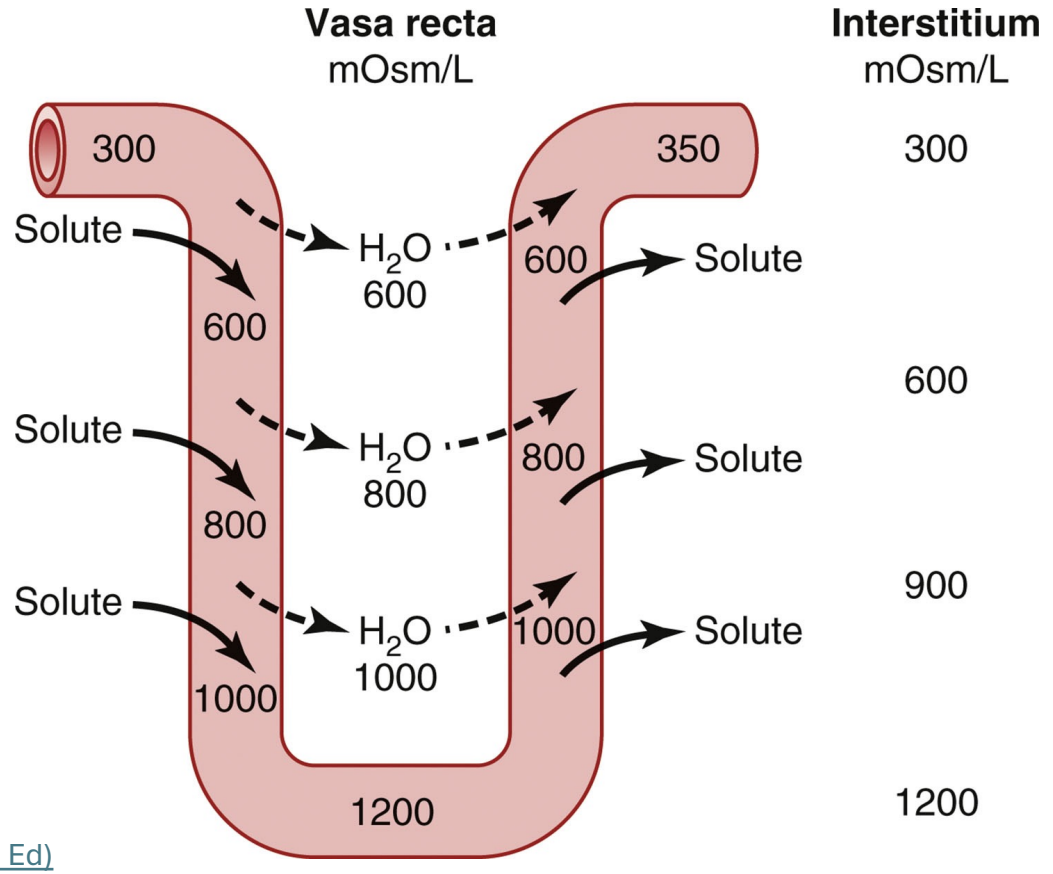
Simple squamous epithelium

- Thin descending
- Thin ascending

Digital Histology

Countercurrent Multiplier & Exchange





Guyton & Hall Textbook of Medical Physiology (15th Ed)

Countercurrent multiplier system

Flow of **tubular fluid** in the nephron loop **establishes osmotic gradient** in the interstitial tissue of the renal pyramids

Countercurrent exchange system

Blood flow in the **vasa recta** runs counter to tubular fluid flow and **maintains the osmotic gradient** while providing blood supply to tissues

Juxtaglomerular Apparatus

Collection of specialized cells at the vascular pole of the glomerulus involved in **regulation of blood pressure**

Macula densa:

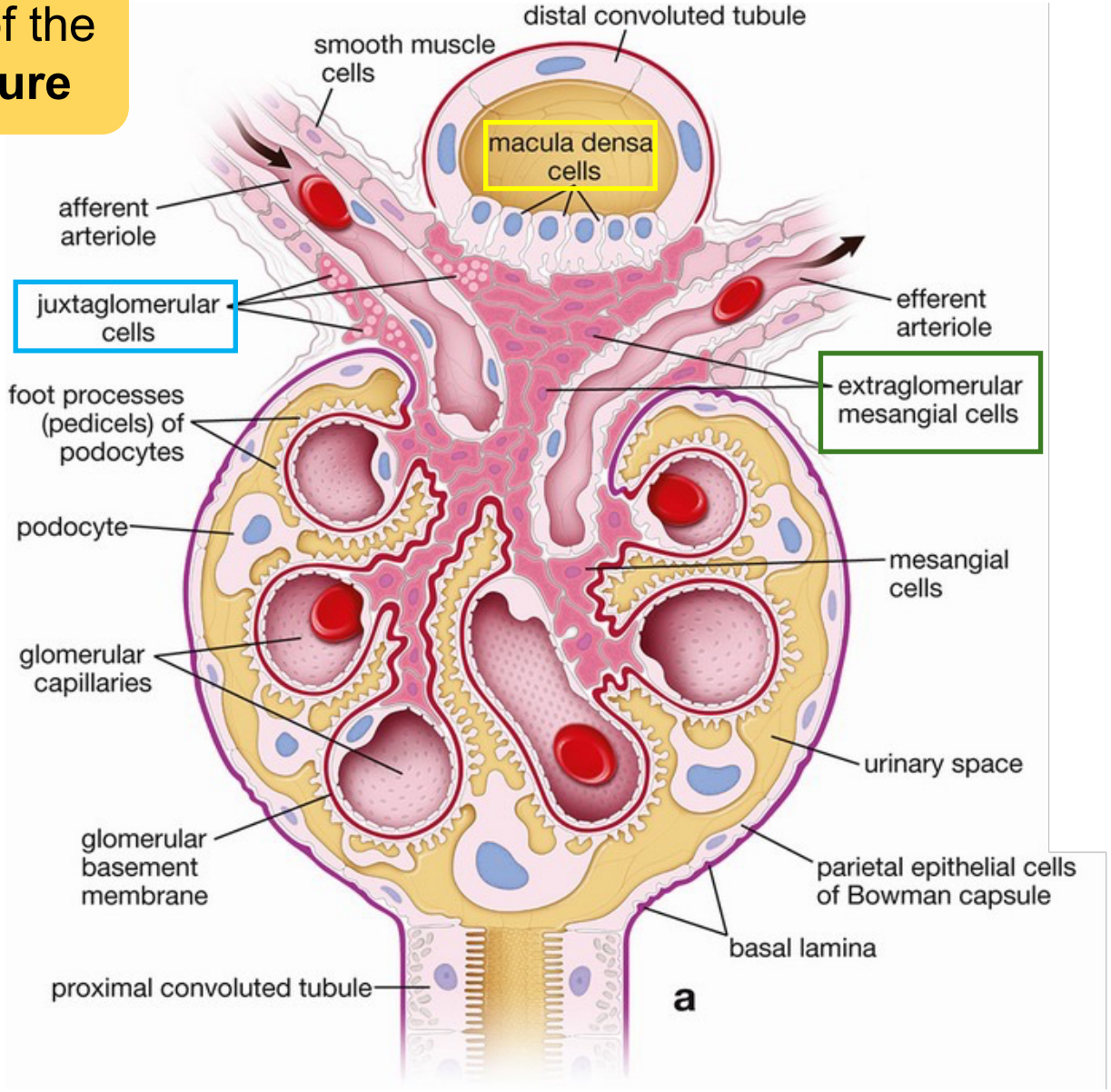
- Specialized columnar epithelial cells at the transition from DST to DCT, adjacent to the vascular pole of the renal corpuscle
- **Monitor luminal Na^+ and Cl^- concentrations** → release vasomotor factors to stimulate arterioles & activate the juxtaglomerular granular cells

Juxtaglomerular granular cells:

- Modified smooth muscle cells of the afferent arteriole
- Secrete **renin** in response to decreased arterial pressure to initiate the renin-angiotensin-aldosterone system (RAAS) that raises blood pressure

Extraglomerular mesangial (lacis) cells:

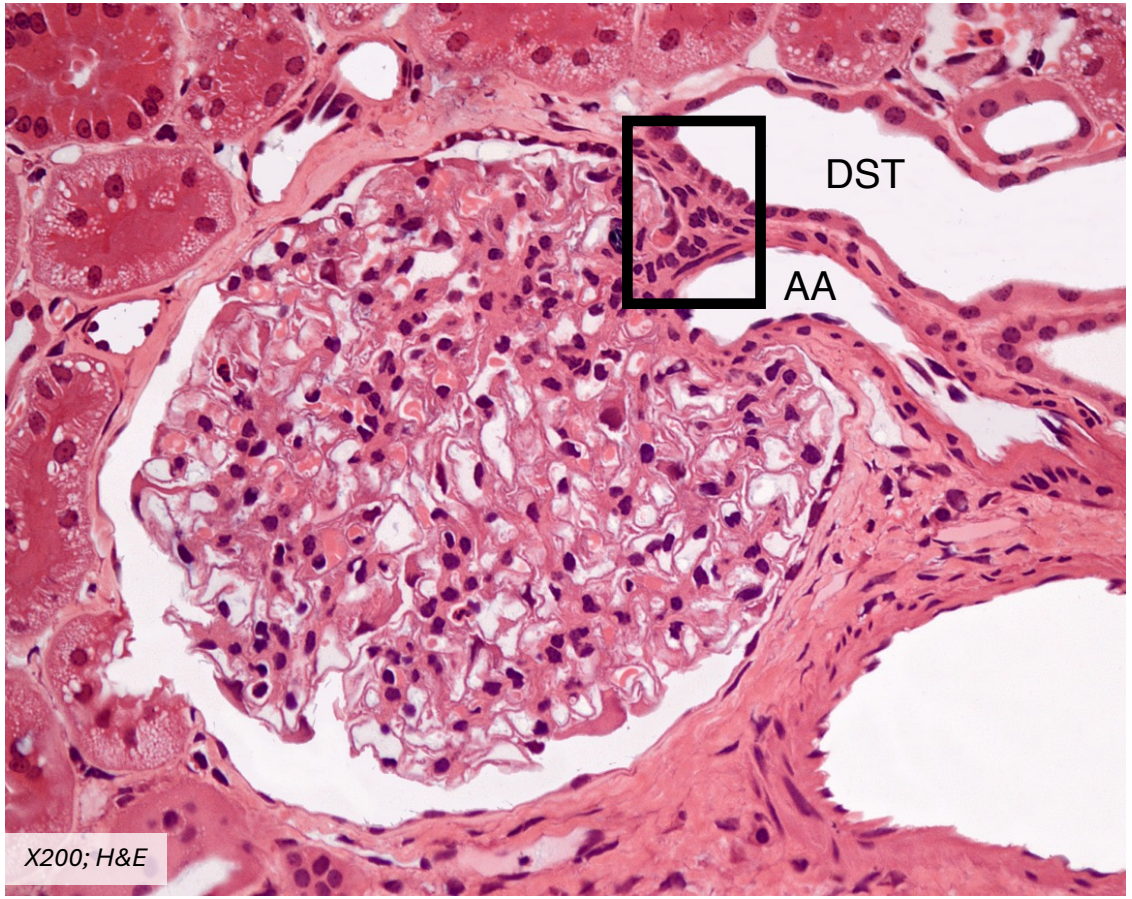
- Similar **supportive, contractile, & defensive functions** as the glomerular mesangial cells



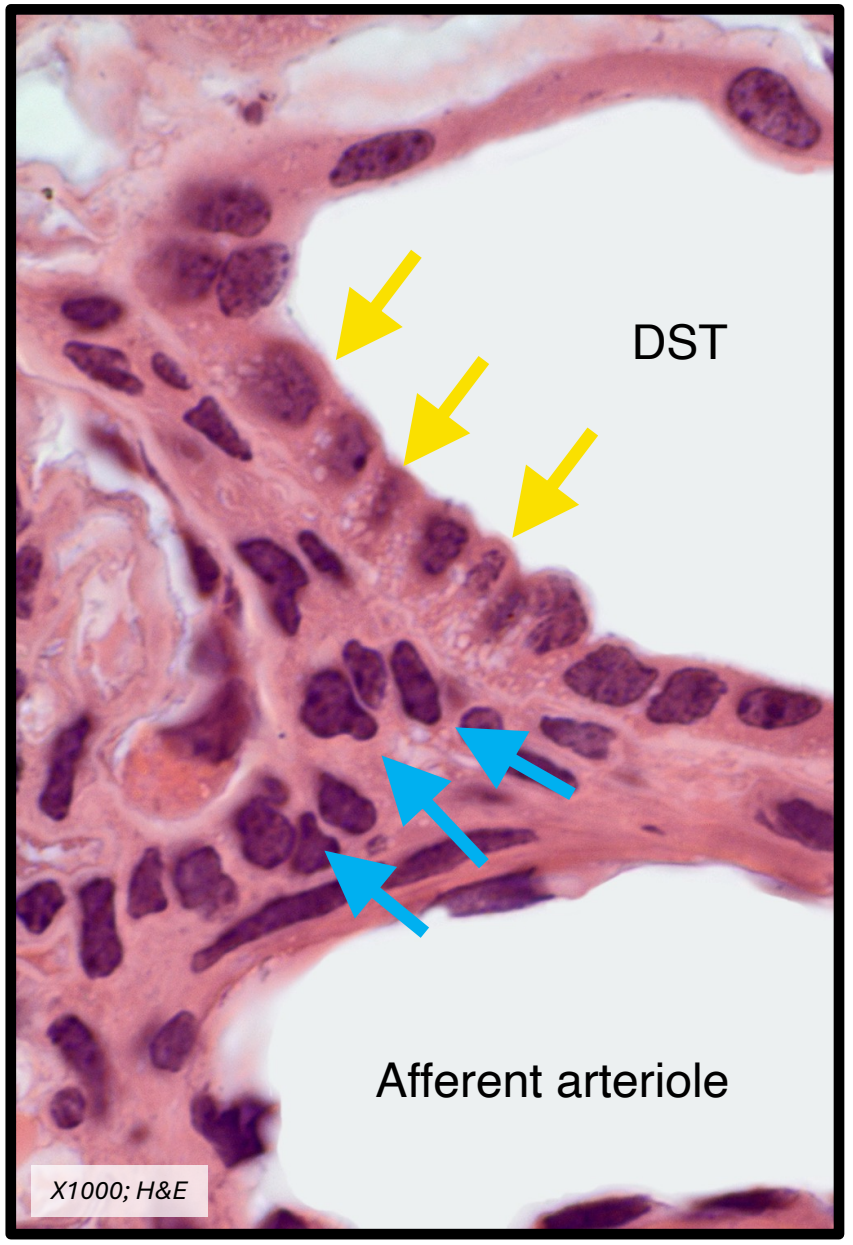
DST – distal straight tubule
AA – afferent arteriole

Juxtaglomerular Apparatus

- **Macula densa**
- **Juxtaglomerular granular cells**



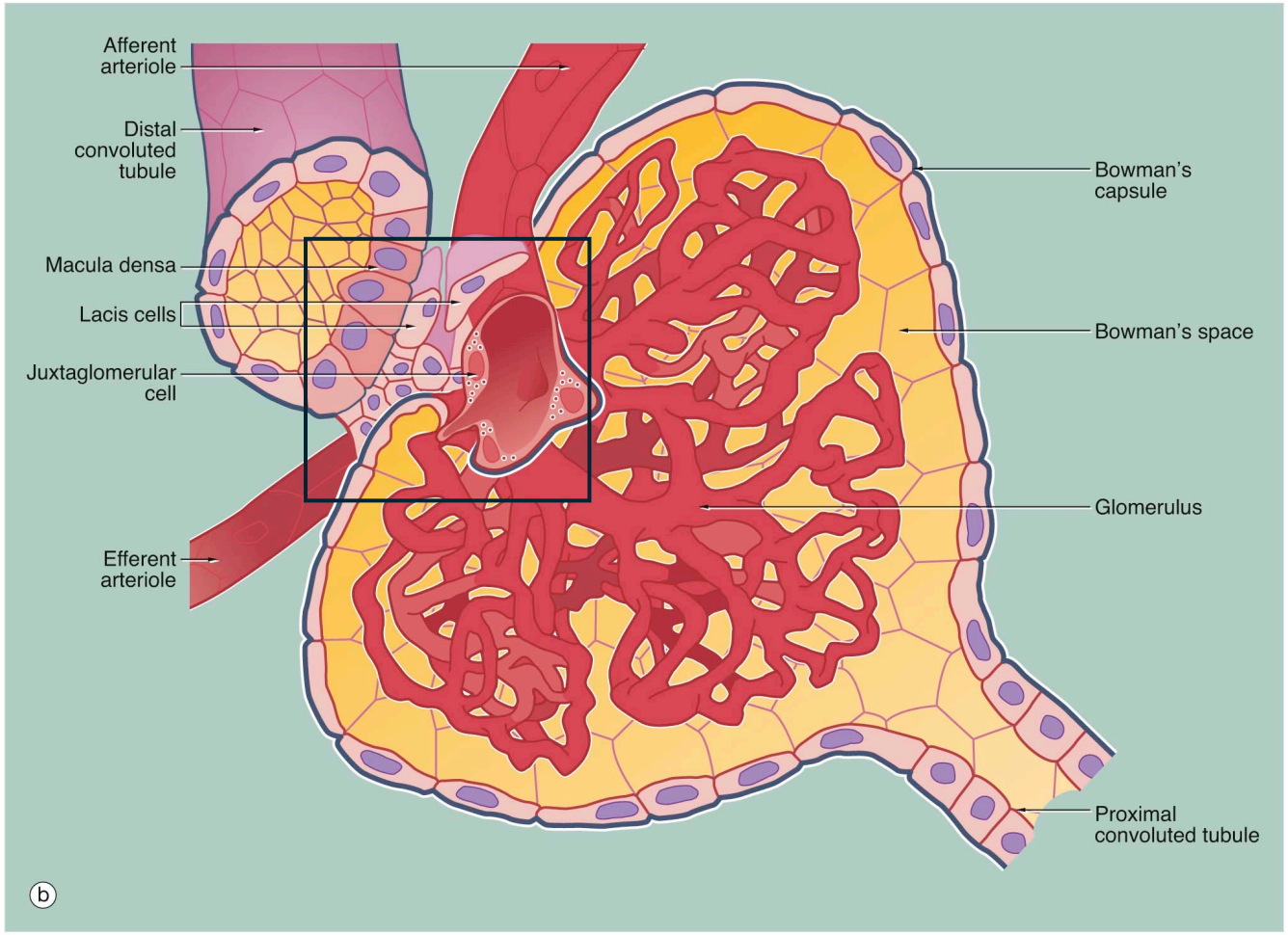
Digital Histology

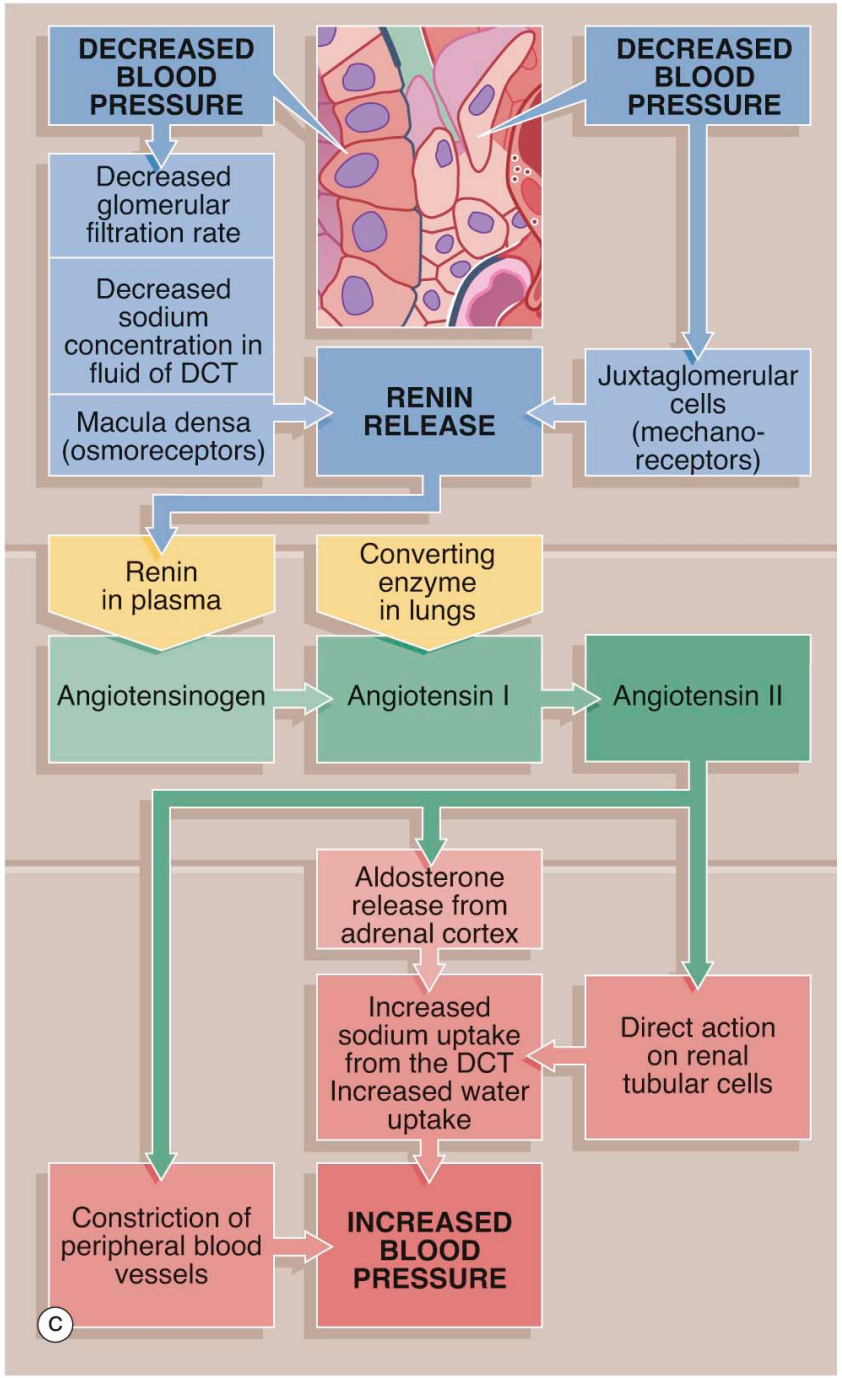


Digital Histology

RAAS

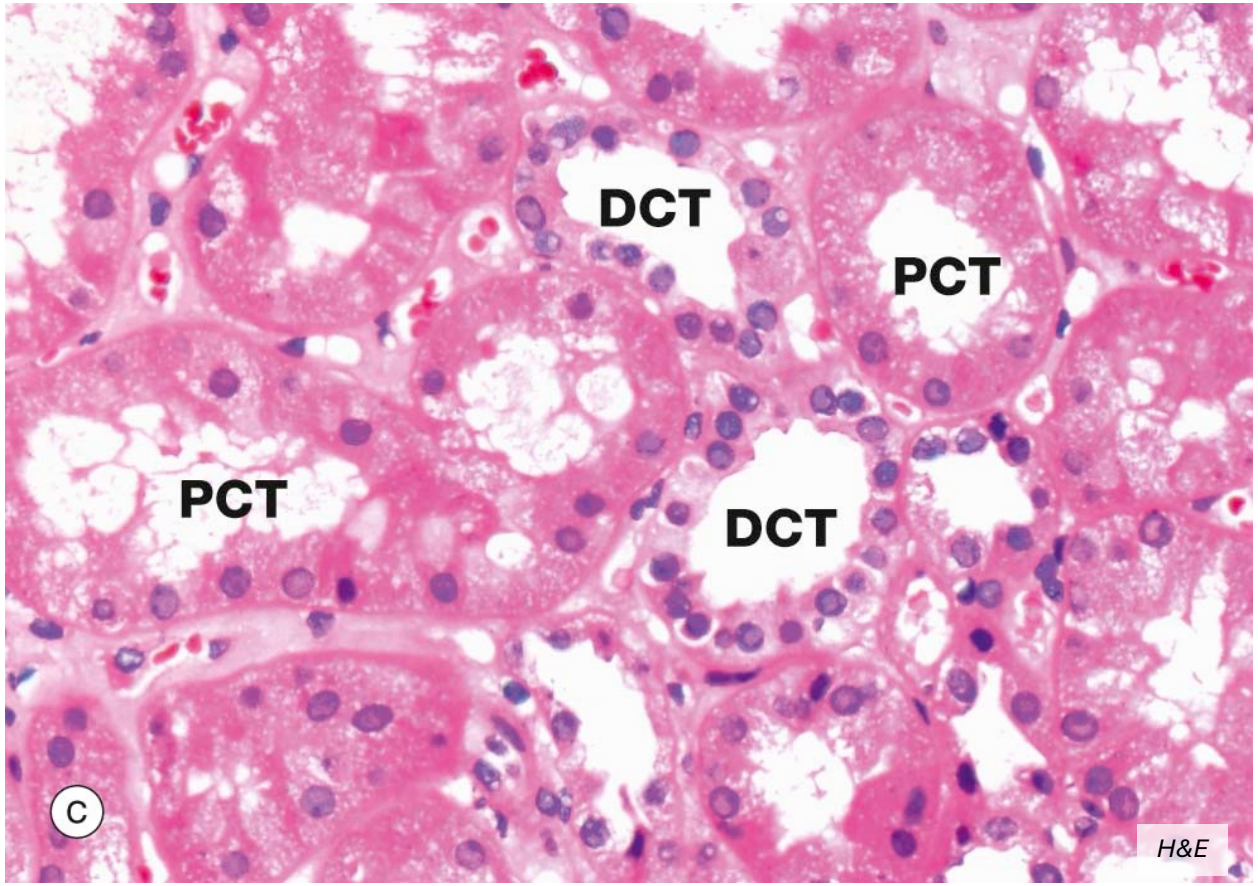
JGA regulates blood pressure by activating the **renin-angiotensin-aldosterone system (RAAS)**





DCT – distal convoluted tubule
PCT – proximal convoluted tubule

Distal Convoluted Tubule



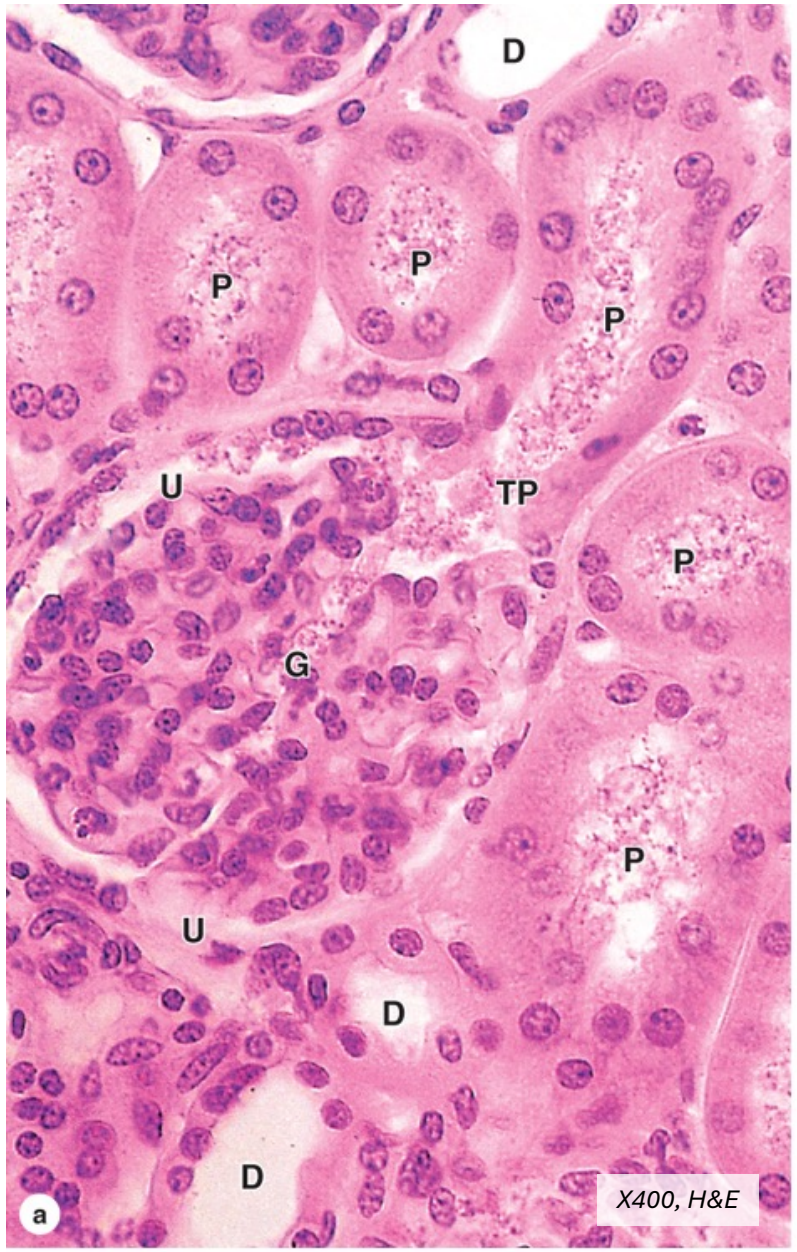
Key functions:

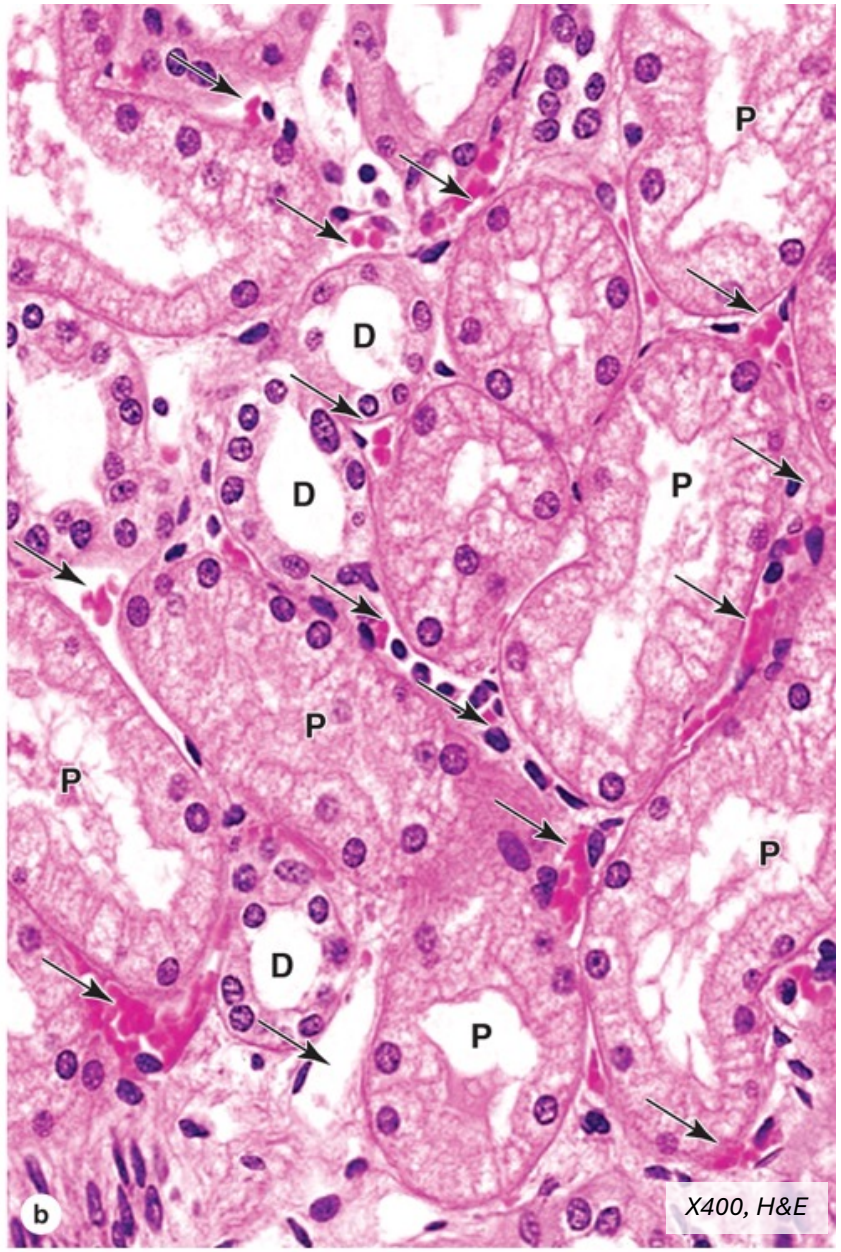
- Reabsorption of Na^+ & H_2O in response to **aldosterone** (from suprarenal glands)
- Ca^{2+} reabsorption in response to PTH (from parathyroid glands)

Key histological features:

- Simple cuboidal epithelium
- Cells smaller than PCT → more nuclei visible and sometimes “bulge” into lumen
- Lack brush border → more “empty” lumen
- Fewer mitochondria than PCT
- Shorter tubule than PCT → fewer cross sections in a routine section

Proximal vs. Distal Convoluted Tubules

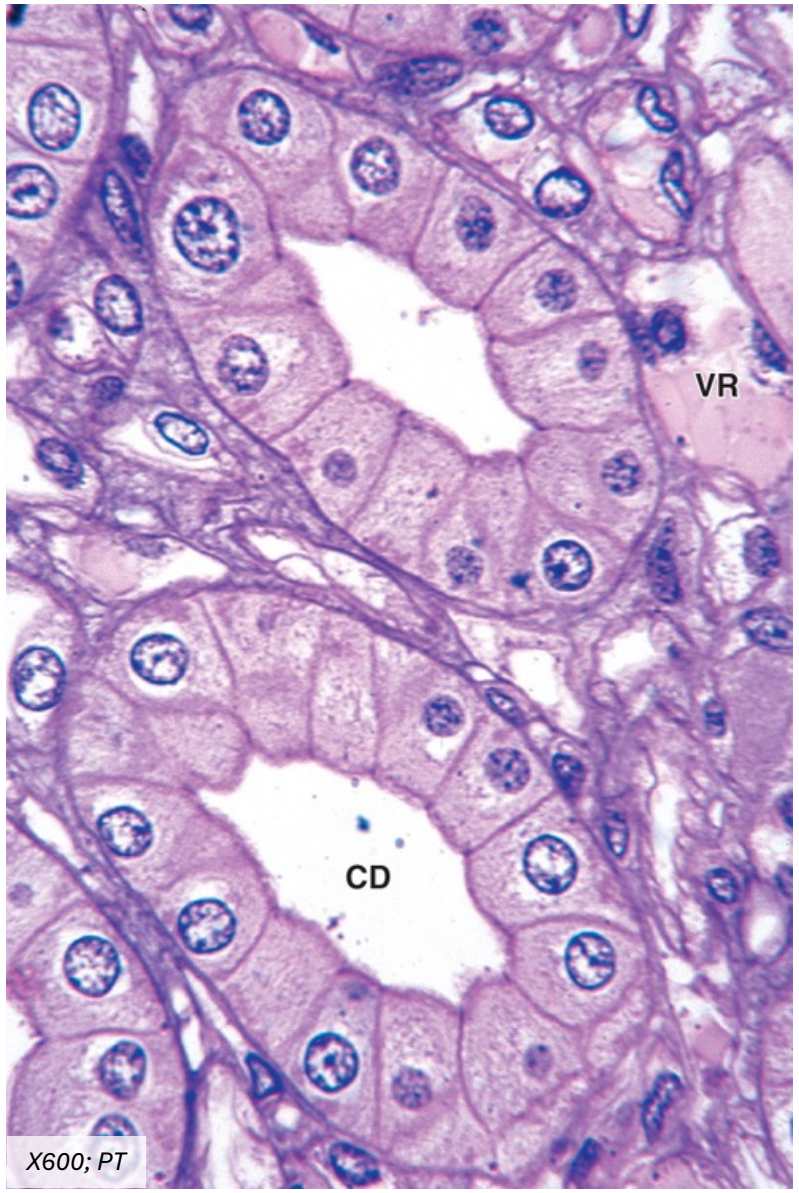




G – glomerulus
U – urinary space
TP - tubular pole of corpuscle
P – proximal convoluted tubule
D – distal convoluted tubule

CD – collecting duct
VR – vasa recta

Collecting Duct



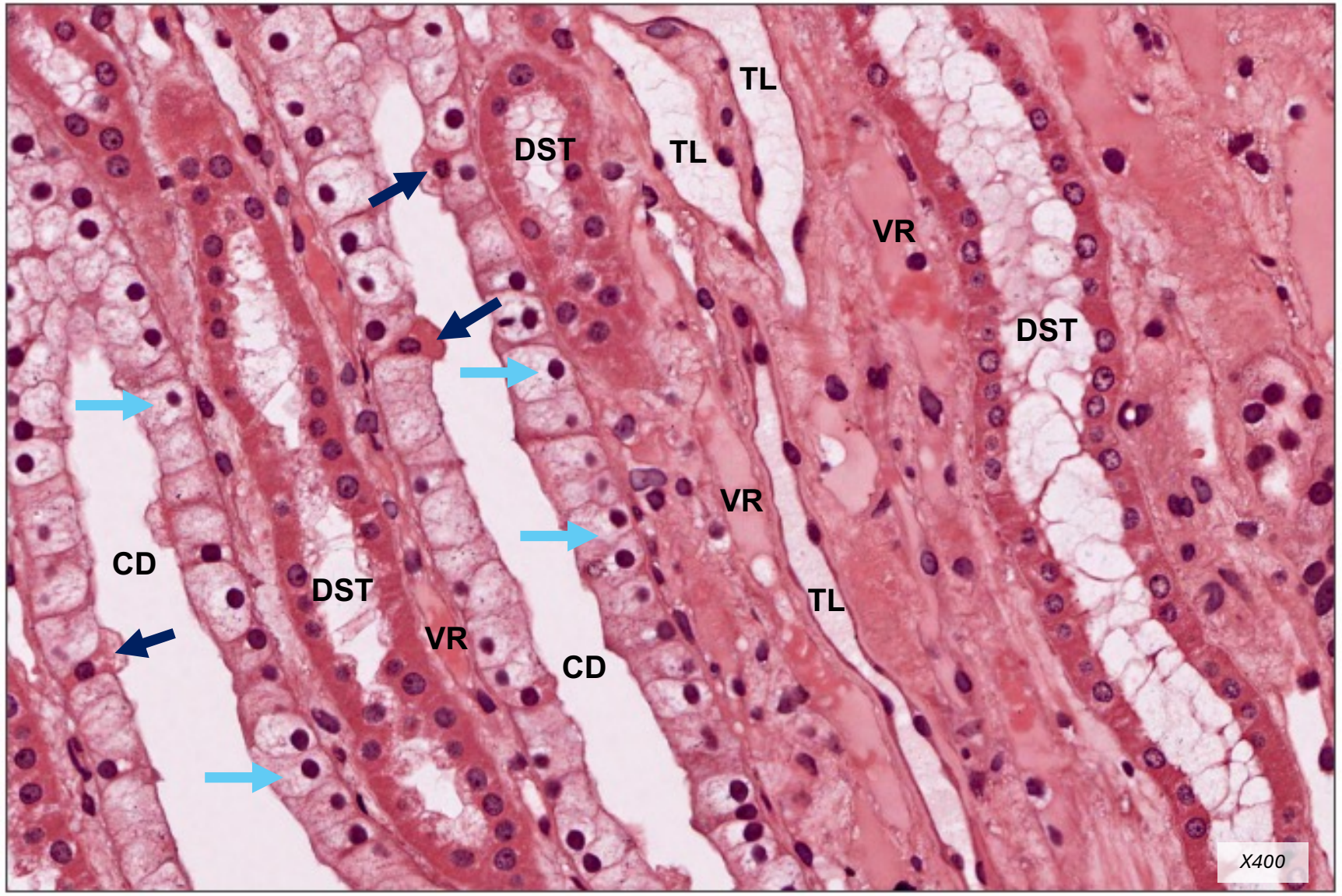
Receives tubular fluid from multiple nephrons & makes final fluid modifications

Key histological & functional features:

- Simple cuboidal epithelium in the cortex → **increasingly columnar** toward the inner medulla
- **Distinct visible lateral borders** in routine section
- Two epithelial cell types:
  - **Principal (light) cells:** most abundant; rich in aquaporin channels for water reabsorption in response to **antidiuretic hormone (ADH)** from posterior pituitary gland
  - **Intercalated (dark) cells:** scattered, visibly darker cells that secrete H^+ (α cells) or HCO_3- (β cells) into tubular fluid to **regulate acid-base balance**

CD – collecting duct
VR – vasa recta
TL – thin limbs of nephron loop
DST – distal straight tubule

Collecting Duct (in Medulla)



Principal cells

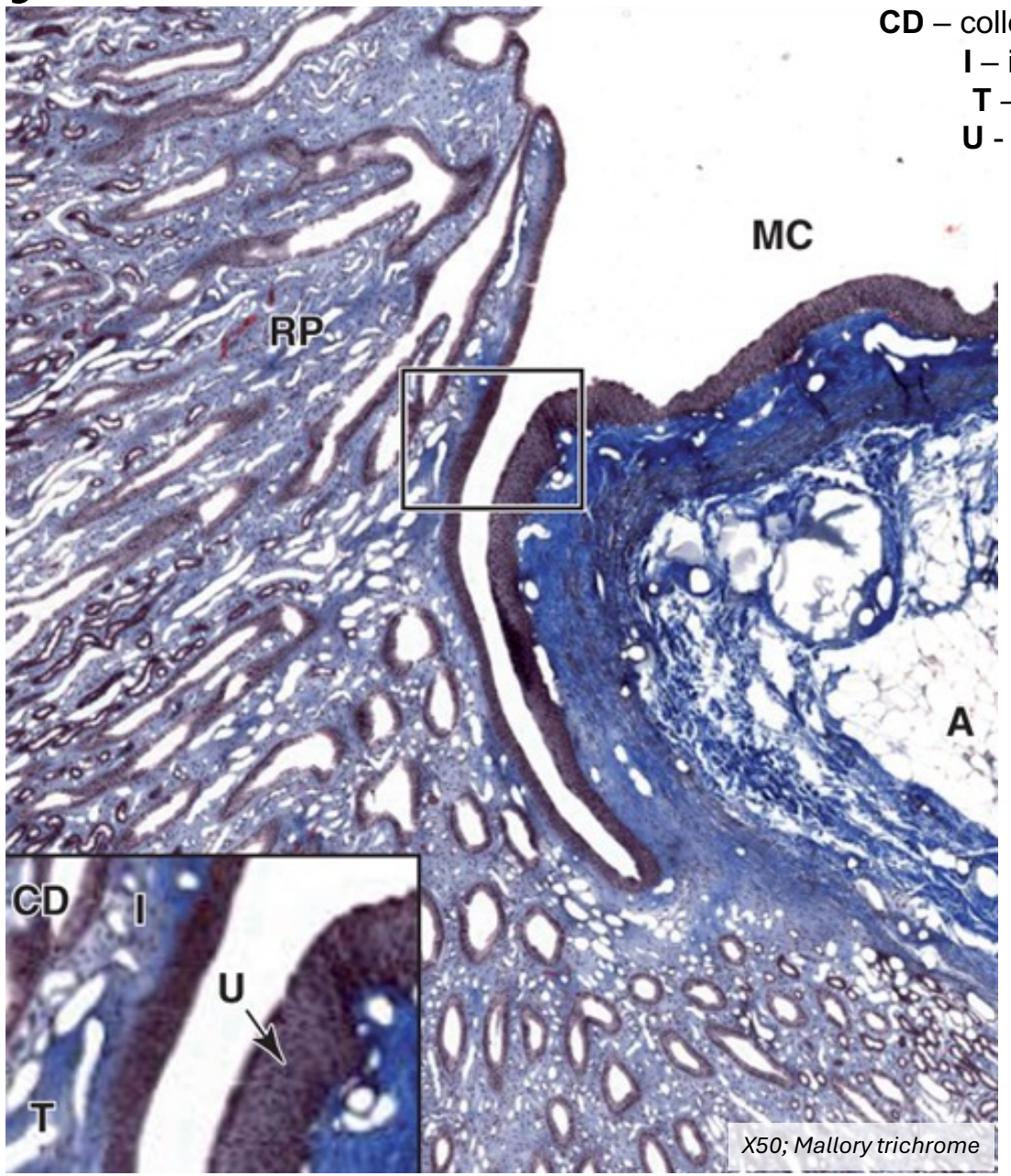
Intercalated cells

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Papillary Ducts

MC – minor calyx
RP – renal papilla
A – adipose
CD – collecting duct
I – interstitium
T – thin limbs
U - urothelium

- Collecting ducts nearing the apex of the renal pyramid converge → form **papillary ducts**
- No modification of tubular fluid (now considered **urine**) in papillary ducts
- Lined by a **simple cuboidal epithelium** (no intercalated cells) that is continuous with the urothelium lining the urinary tract
- Papillary ducts drain at the **area cribrosa** of the renal papilla (apex of the renal pyramid) into the major calyx



Tubules in the Cortical Labyrinth

Proximal Convoluted Tubules

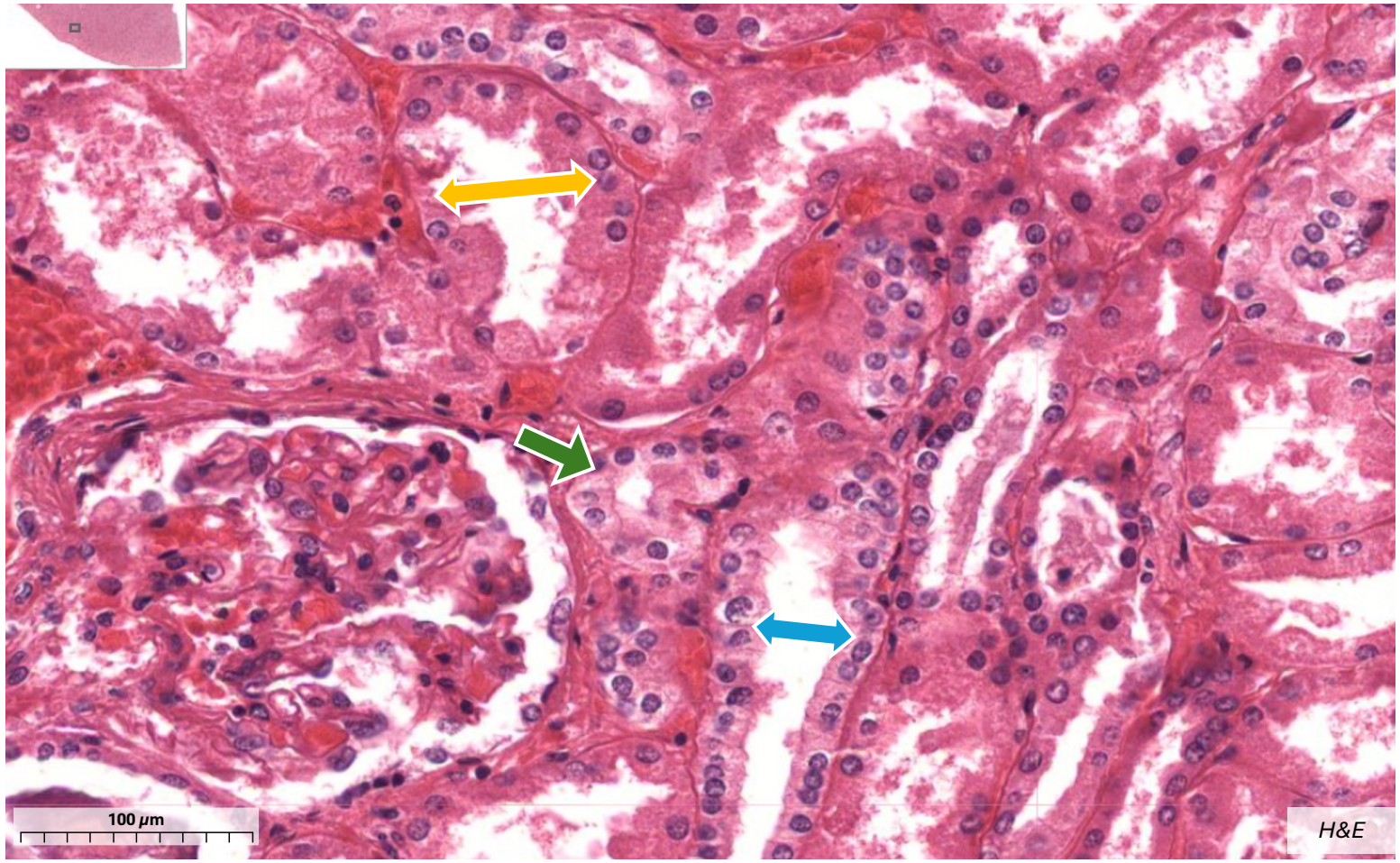
- **Brush border** → irregular lumen
- Large, irregularly spaced nuclei
- No clear boundary between cells
- More acidophilic than DCT
- More numerous profiles than DCT

Distal Convoluted Tubules

- **No brush border** → regular lumen
- No clear boundary between cells
- More neutral staining than PCT
- Fewer profiles than PCT

Collecting Duct*

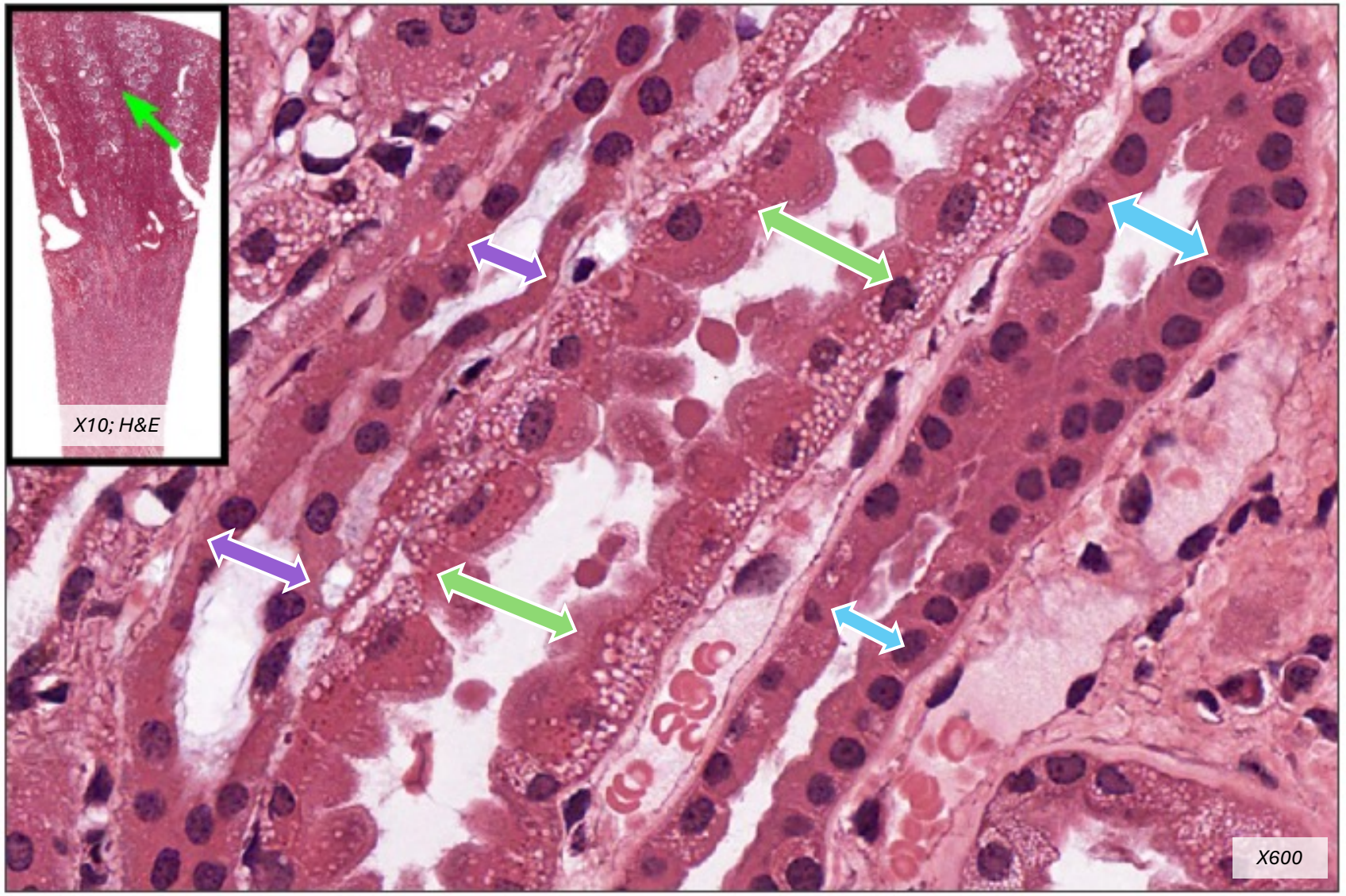
- No brush border → regular lumen
- **Well-defined cell boundaries**
- Lightly stained principal cells (& few scattered darker-stained intercalated cells)



HistologyGuide

*Collecting duct runs within the medullary ray, but portions are often visible in sections showing the cortical labyrinths

Tubules in the Medullary Ray



Digital Histology

Collecting Duct

- Cuboidal epithelium (becomes more columnar in the medulla)
- **Conspicuous cell boundaries**

PST

- Taller cuboidal epithelium than DST, with **brush border**
  (partly obscured by luminal contents)

DST

- Shorter cuboidal epithelium
- Irregularly spaced nuclei that often bulge into lumen

Note: the “foamy” appearance of the PST cells in this section is due to poor fixation of mitochondria

Summary

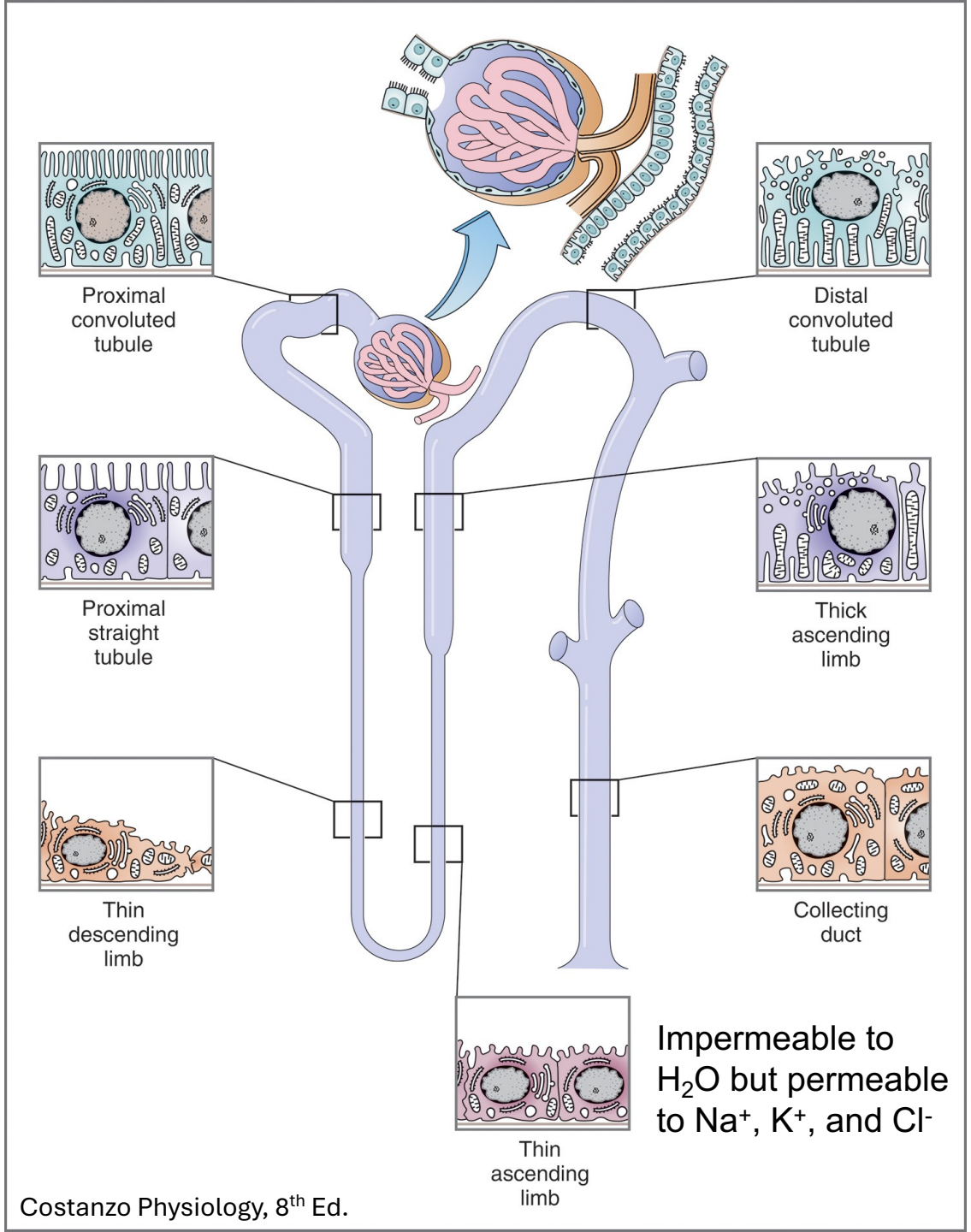
Region	Location	Histologic Features	Major Functions
Glomerular capsule	Cortex (cortical labyrinth)	Simple squamous epithelium (parietal layer) and podocytes (visceral layer)	Surrounds glomerulus, filters blood
Proximal convoluted tubule	Cortex (cortical labyrinth)	Simple cuboidal epithelium with brush border	Reabsorbs all glucose and amino acids, majority of Na+, Cl-, and H2O
Thick descending limb / proximal straight tubule	Cortex (medullary ray), Medulla (OSOM)	Simple cuboidal epithelium with brush border	Similar to PCT
Thin descending limb	Medulla (ISOM & Inner)	Simple squamous epithelium	Permeable to H2O (reabsorption)
Thin ascending limb	Medulla (Inner)	Simple squamous epithelium	Impermeable to H2O, but permeable to Na+, K+, and Cl-
Thick ascending limb / distal straight tubule	Medulla (ISOM & OSOM), Cortex (medullary ray)	Simple cuboidal epithelium without brush border	Reabsorption of Na+ through active transport
JGA: Macula densa	Cortex (distal straight tubule)	Simple cuboidal epithelium	Monitors luminal Na+ and Cl- levels
JGA: Juxtaglomerular granular cells	Cortex (afferent arteriole)	Modified smooth muscle cells	Synthesize renin & release it into bloodstream
Distal convoluted tubule	Cortex (cortical labyrinth)	Simple cuboidal epithelium	Reabsorption of Na+ and H2O in presence of aldosterone
Collecting ducts	Cortex (medullary ray) & Medulla (all parts)	Simple cuboidal epithelium (cortex); simple columnar epithelium (medulla)	Reabsorption of H2O in presence of antidiuretic hormone (ADH)

Summary of Tubule Functions

Reabsorbs all glucose and amino acids, majority of Na^+, Cl^-, and H_2O

Reabsorbs remaining glucose

Permeable to H_2O



Reabsorption of Na^+ and H_2O in response to aldosterone

Reabsorption of Na^+ through active transport

Reabsorption of H_2O in response to antidiuretic hormone (ADH)

Impermeable to H_2O but permeable to Na^+, K^+, and Cl^-

Costanzo Physiology, 8th Ed.

Pause & Recall: Renal Corpuscle

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

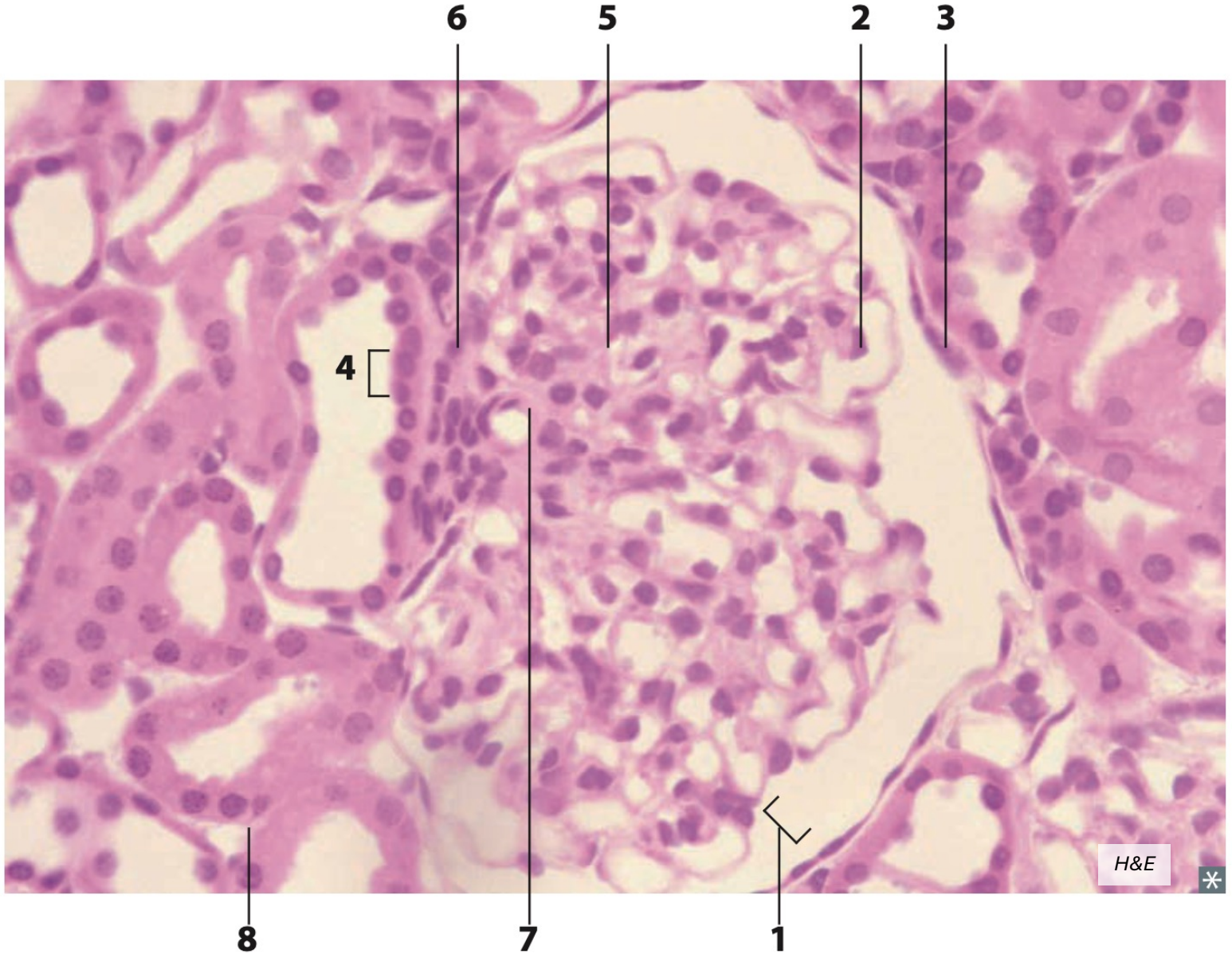
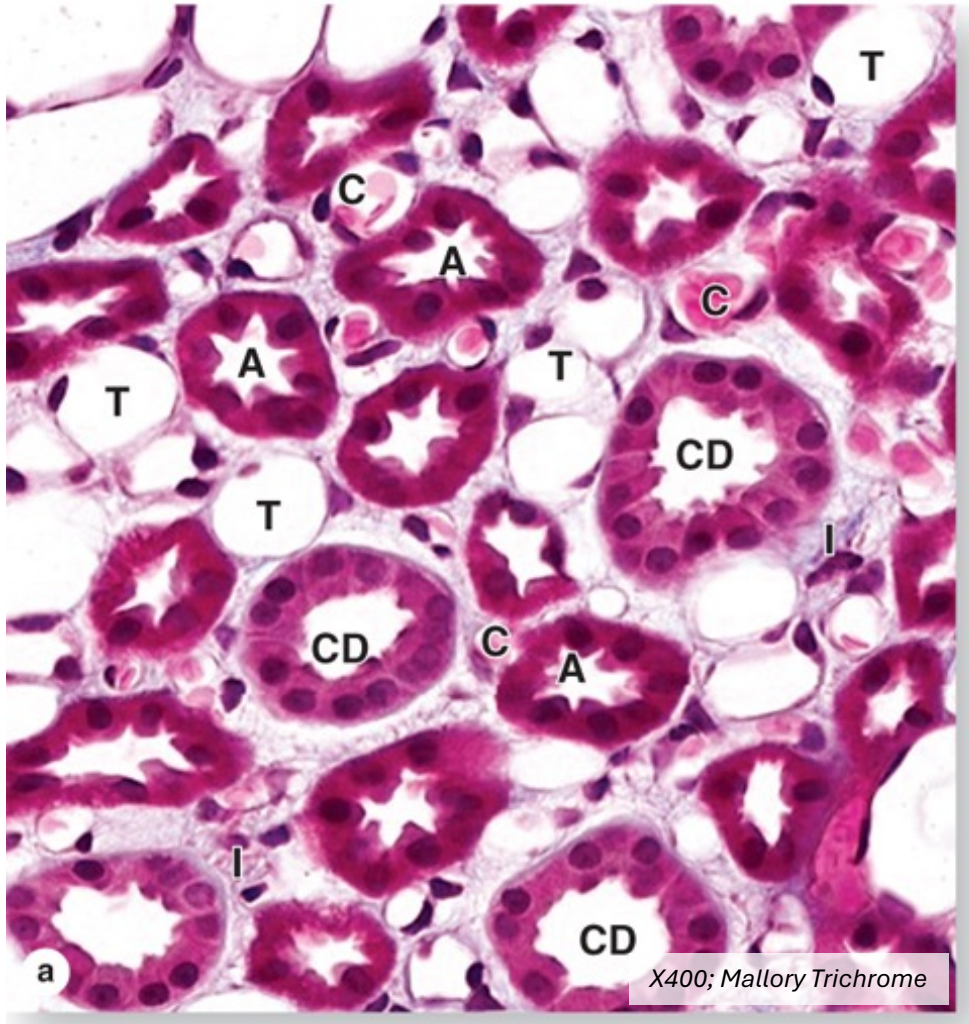


Image: FirstAid for the Basic Sciences (Organ Systems), 3rd Ed.

T – thin descending limb
A – thick ascending limb
C – vasa recta capillaries
CD – collecting ducts
I – interstitium

Nephron Loop

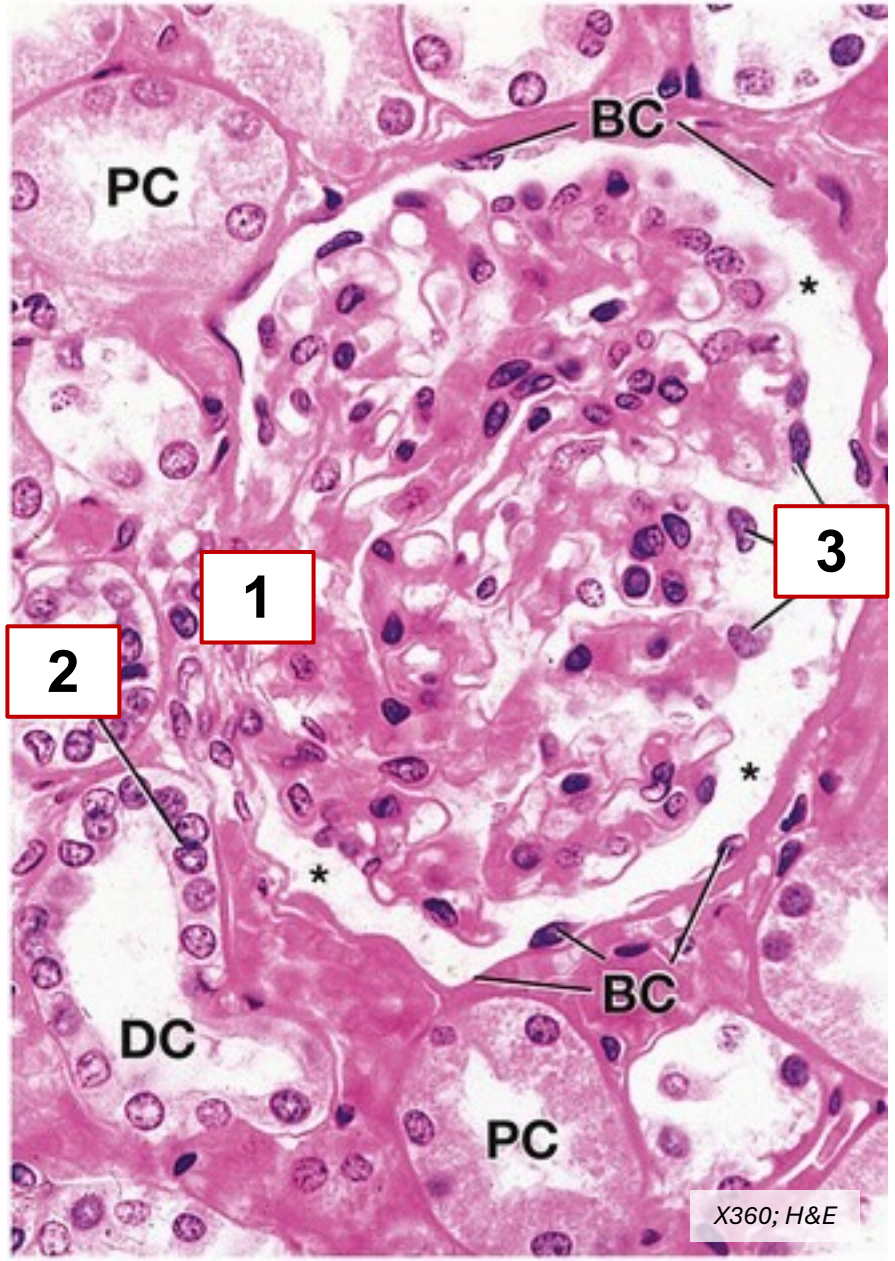




Which layer of the medulla is shown in the left-hand slide?

PC – proximal convoluted tubule
DC – **d**istal convoluted tubule
BC – parietal layer of Bowman's capsule

Pause & Recall

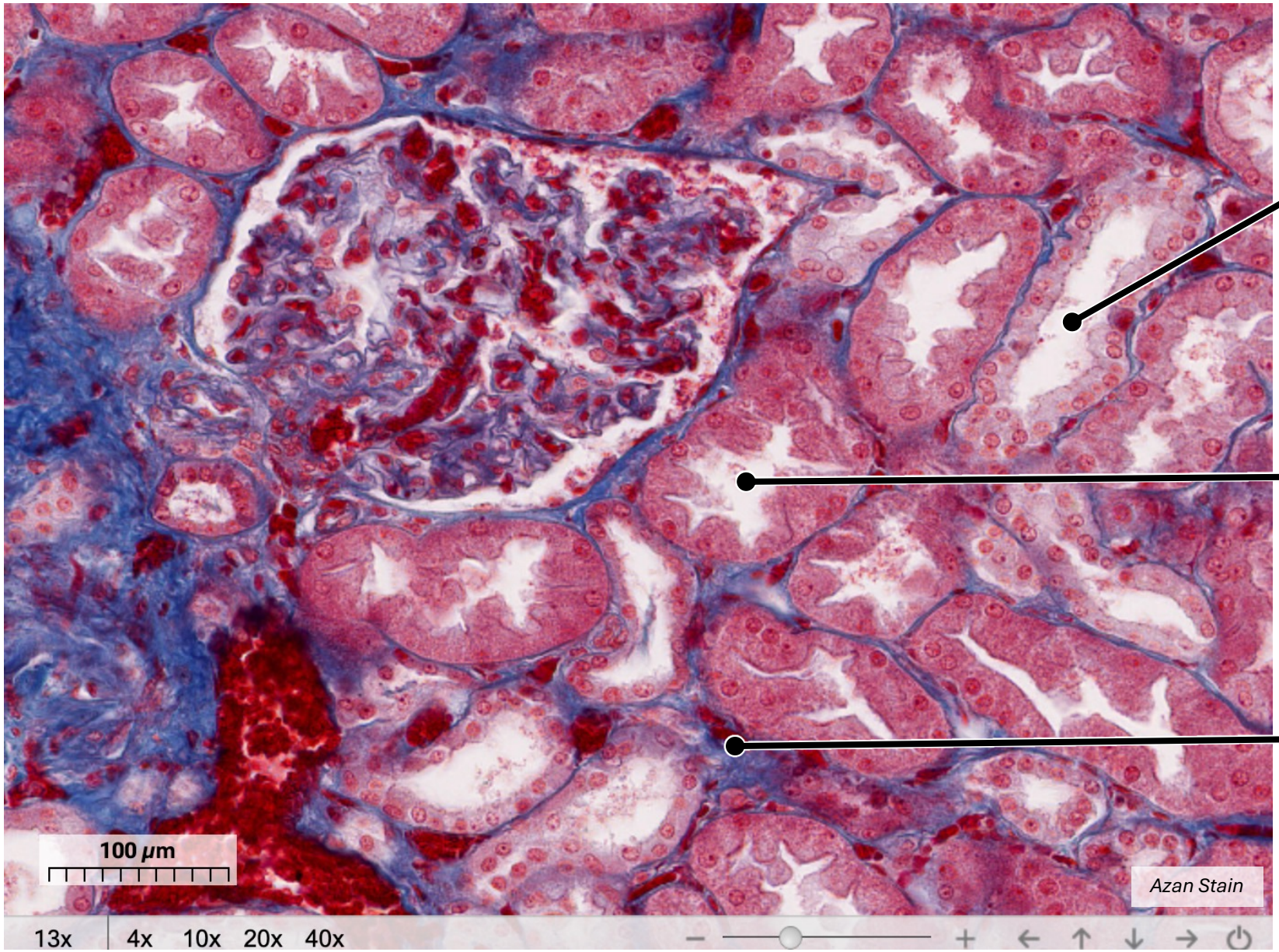


What region of the renal corpuscle is indicated by #1?

What is the function of the cells indicated by #2?

Interdigitation of the cells indicated by #3 forms what part of the glomerular filtration barrier?

Pause & Recall

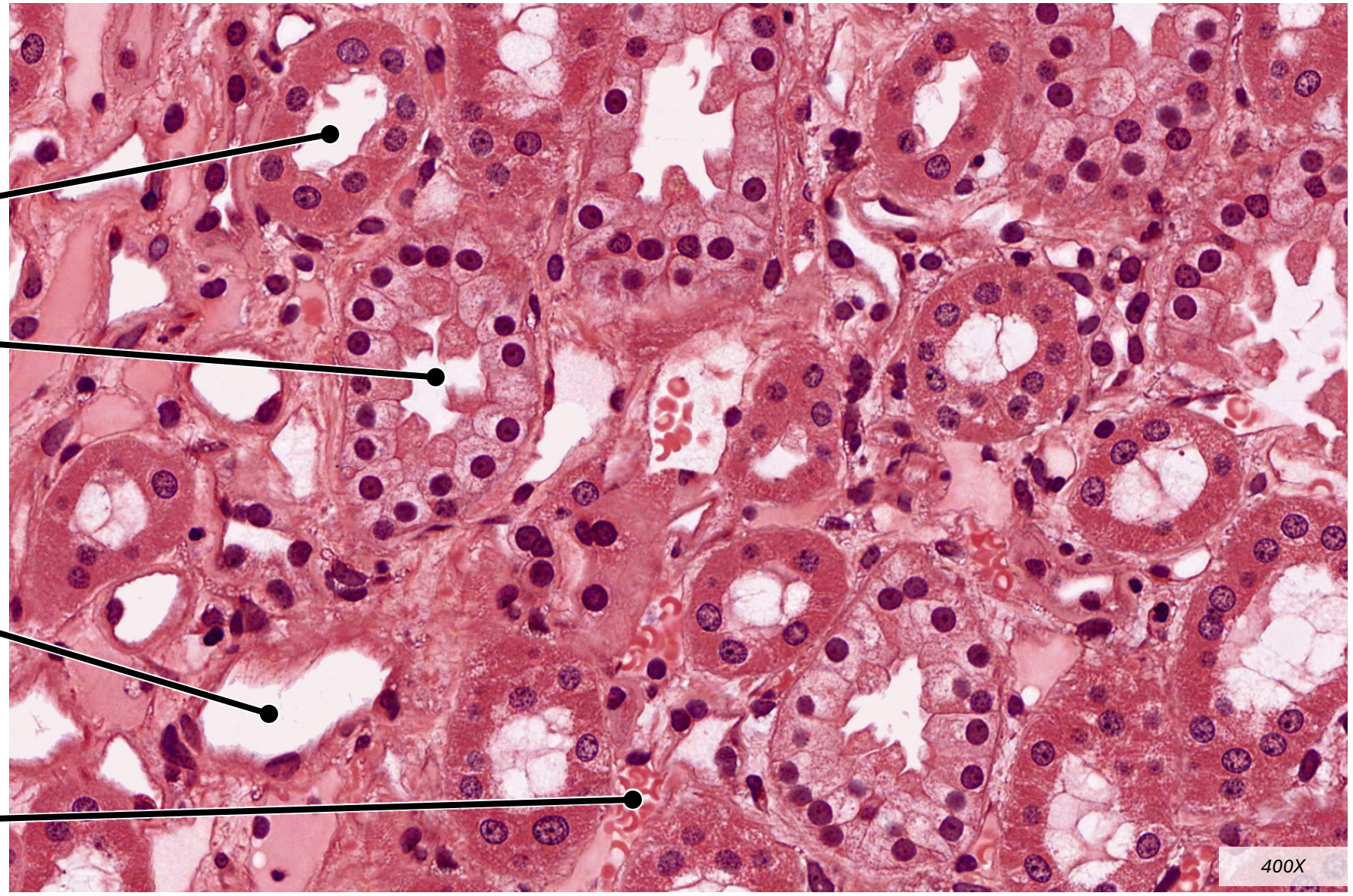


Identify the structure:

Histology Guide

Pause & Recall

In this ISOM section, identify:



Digital Histology

ANSWER KEYS

Pause & Recall: Renal Corpuscle

1. Capsular space
2. Visceral capsule layer
3. Parietal capsule layer
4. Macula densa
5. Glomerular mesangial cell
6. Extraglomerular mesangial cell
7. Glomerular arteriole
8. Peritubular capillary

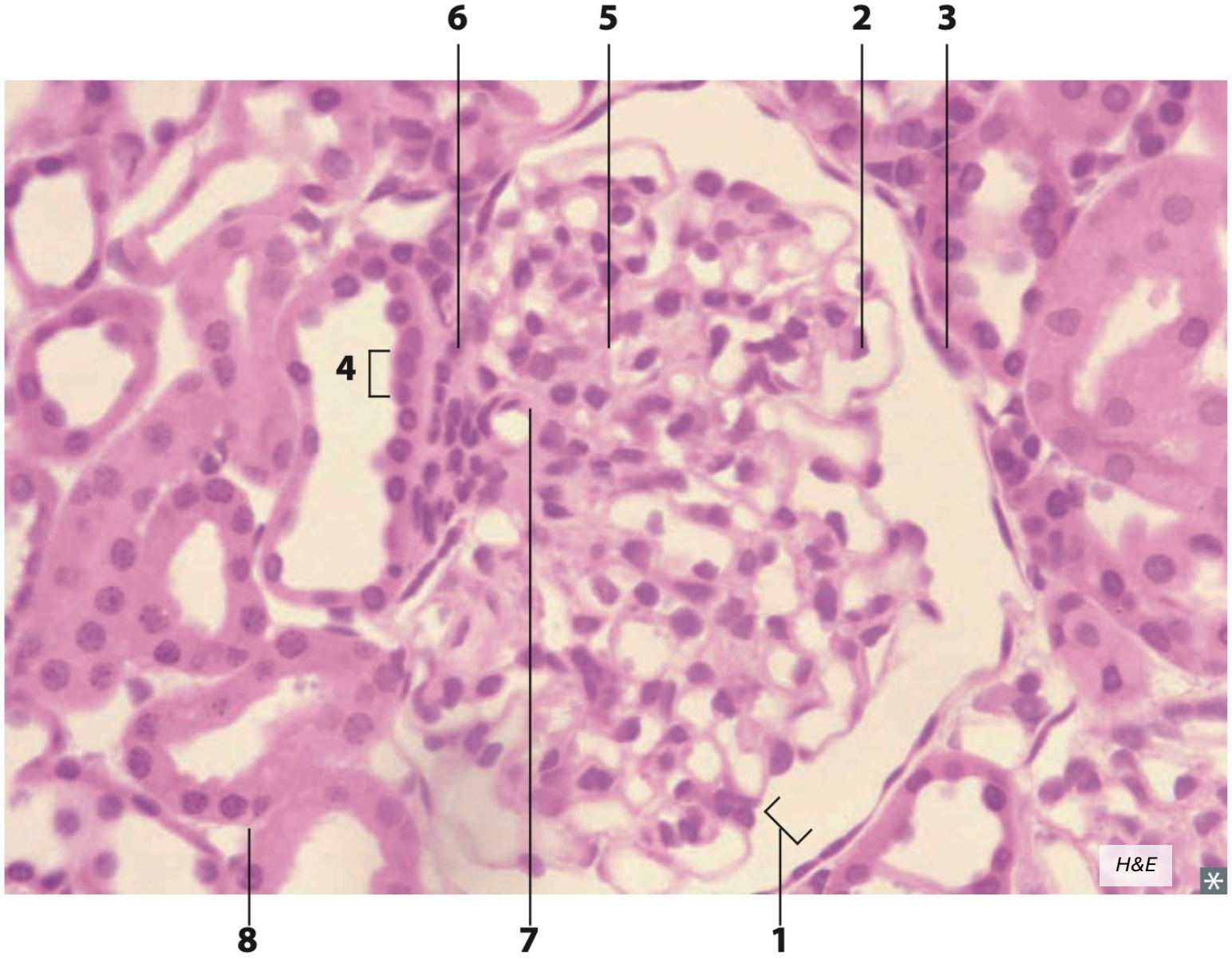
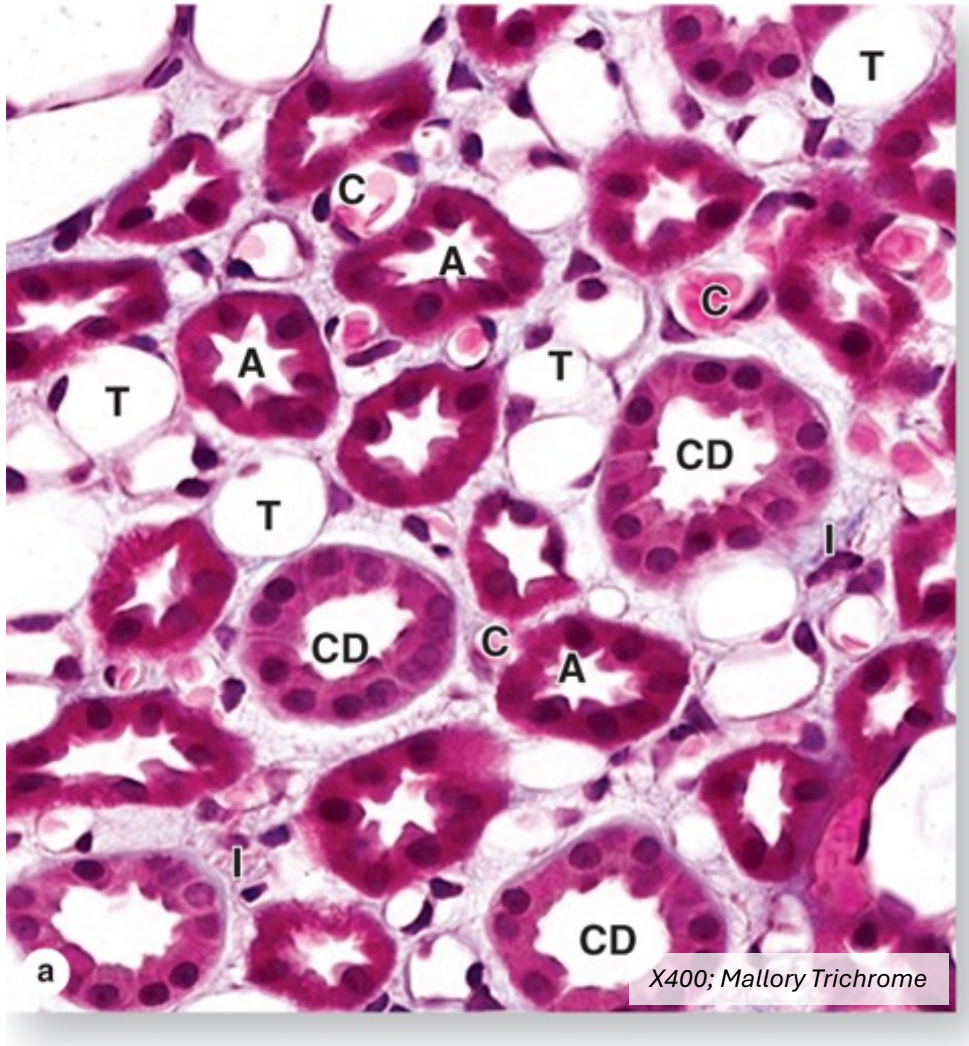
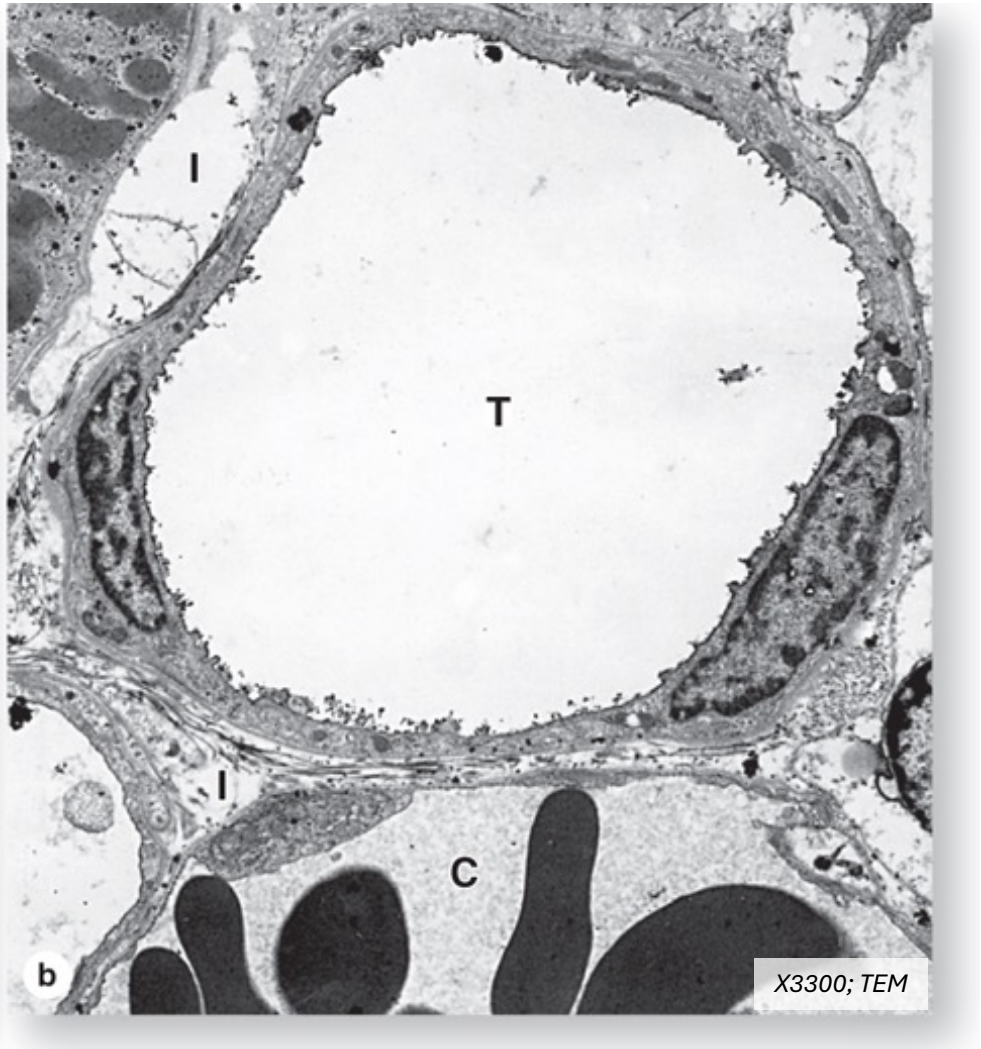


Image: FirstAid for the Basic Sciences (Organ Systems), 3rd Ed.

T – thin descending limb
A – thick ascending limb
C – vasa recta capillaries
CD – collecting ducts
I – interstitium

Nephron Loop

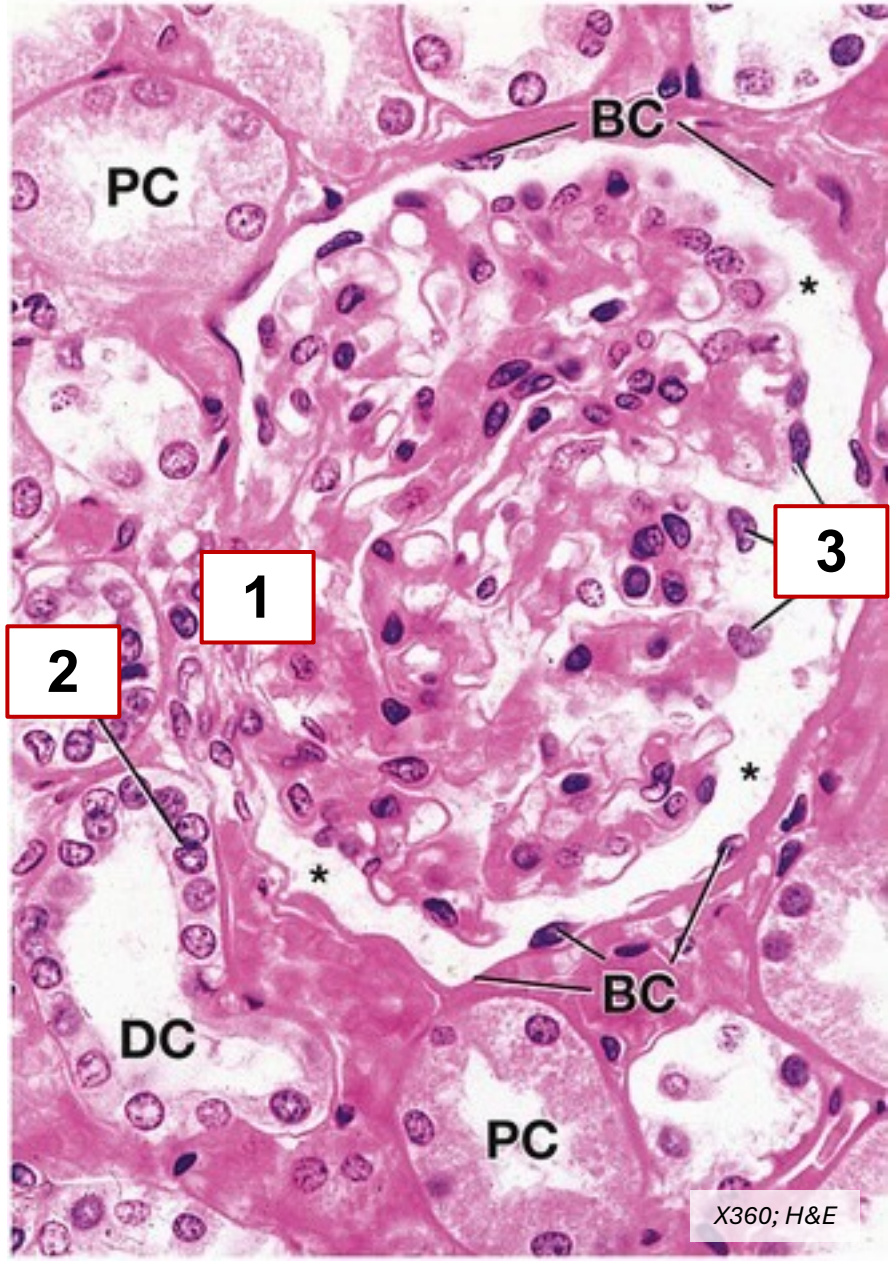




Which layer of the medulla is shown in the left-hand slide? Inner Stripe of the Outer Medulla (ISOM)

PC – proximal convoluted tubule
DC – **d**istal convoluted tubule
BC – parietal layer of Bowman's capsule

Pause & Recall



What region of the renal corpuscle is indicated by #1?

The vascular pole

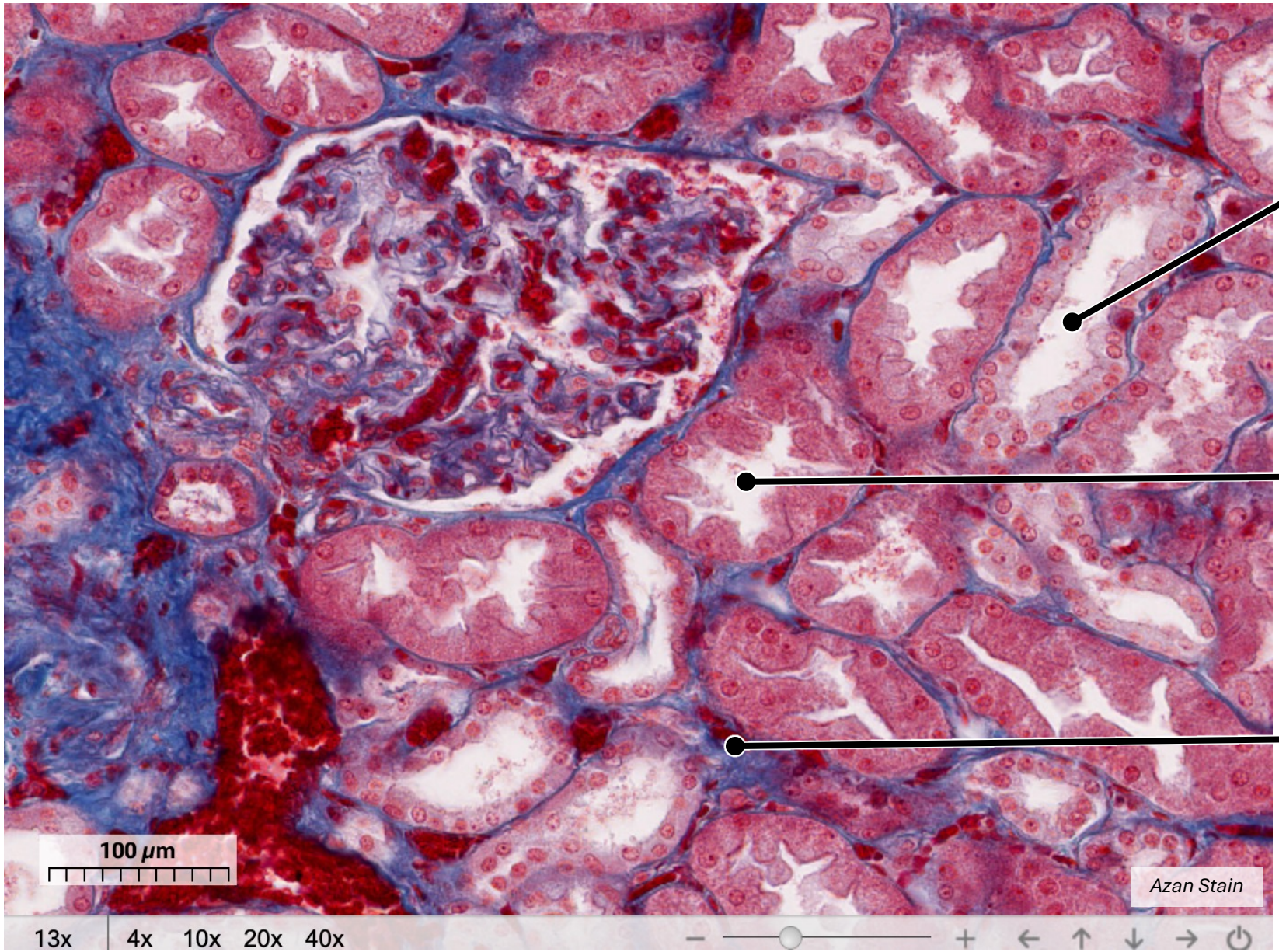
What is the function of the cells indicated by #2?

Monitor Na+ and Cl- concentrations in tubular fluid (macula densa)

Interdigitation of the cells indicated by #3 forms what part of the glomerular filtration barrier?

Filtration slits (podocytes)

Pause & Recall



Identify the structure:

Distal convoluted tubule

Proximal convoluted tubule

Peritubular capillary

Histology Guide

Pause & Recall

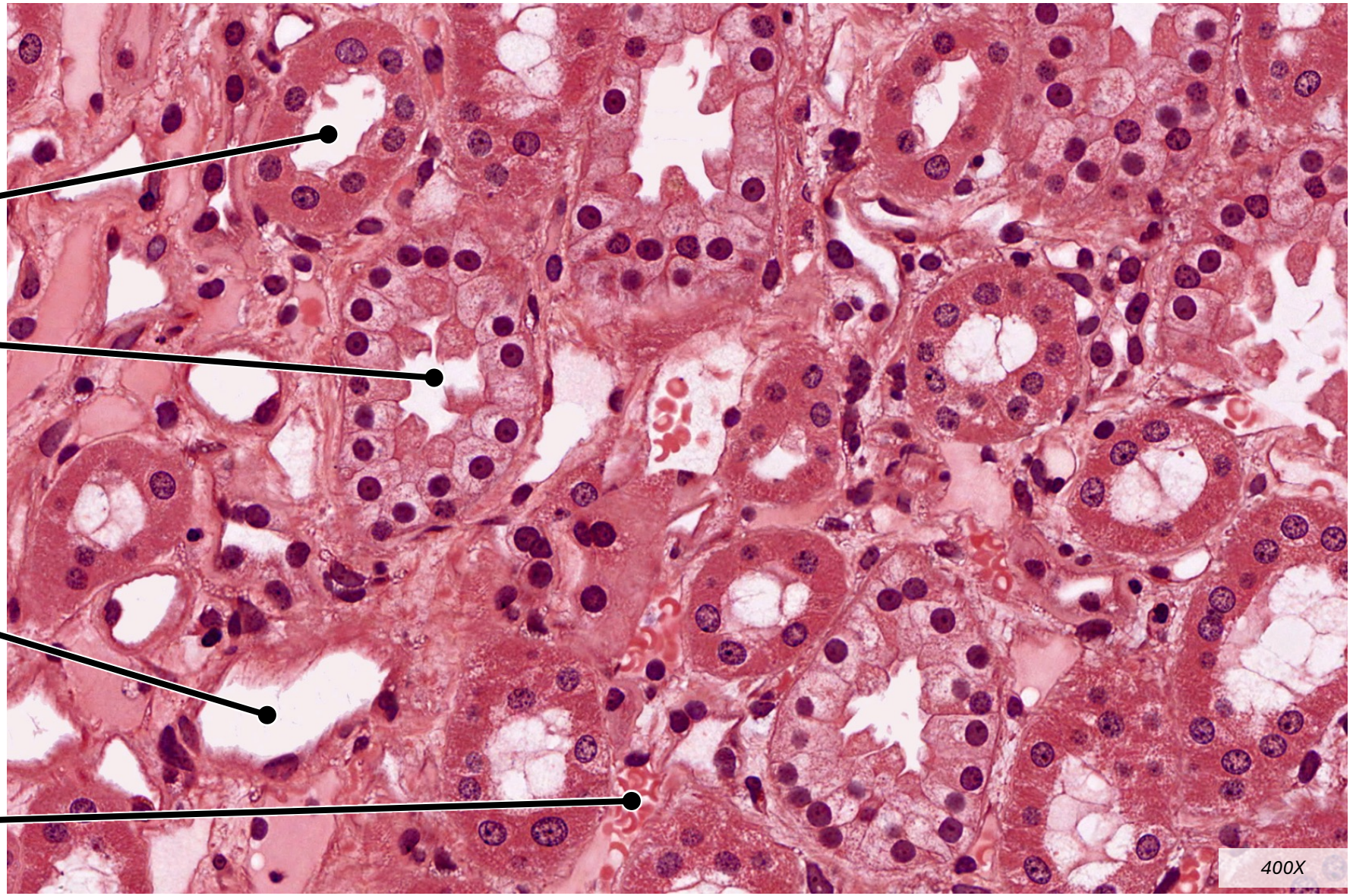
In this ISOM section, identify:

Thick ascending limb

Collecting duct

Thin descending limb

Vasa recta capillary



Digital Histology